Reproducible Pipeline for Juvenile Salmonid Abundance: Subbasin Level

1. Introduction

Several populations of Columbia River Basin Chinook Salmon and Steelhead are endangered or threatened, and therefore are constantly monitored for population recovery efforts (Ford et al., 2025). Data on juvenile abundance are central for these efforts, since it underlies our understanding of population dynamics, including density-dependent effects (Zabel et al., 2006) and the impact of hatchery release practices on natural origin populations (Kostow, 2012).

The majority of the data currently available on wild juveniles, hatchery releases, and spawner abundance are on a fine scale including individual passive integrated transponder (PIT) tagged fish, or tributary-level counts and estimates. This level of detail is well suited for studies of individual, small populations, but basin scale studies of salmon life history and population dynamics models require broader indices of abundances, which are not readily available (Jaeger & Scheuerell, 2023). The finer scale data that does exist is spread across multiple sources with different levels of aggregation, data coverage, and recording/capture methods, and there is currently no process for linking or aggregating these sources together.

This project was a first step towards addressing that gap. Before broadscale abundance estimates could be built, it was necessary to take stock of the finer-scale data that already exists, identify the highest quality sources with the most comprehensive coverage, and aggregate that data to the same level of detail. We developed a reproducible aggregation pipeline that collates subbasin level data summarizing wild juvenile outmigrants, hatchery releases, and spawner abundance estimates into a single data pipeline. This data pipeline can be used to calculate productivity measures such as juvenile outmigrates per spawner, assess drivers of population dynamics, and support future work aimed at generating larger scale abundance estimates.

2. Methods

2.1 Data Types and Sources

The data utilized within this project falls under three broad categories including wild juvenile outmigrant abundance, hatchery origin juvenile releases, and adult spawner abundance (both natural and total). Juvenile Outmigrant abundances (JuvOut) come directly from StreamNet High Level Indicators (HLI) tables that standardize and review abundance data according to population IDs defined by NMFS (StreamNet, 2021a). Recruits per Spawner (RperS) and Naturally Occurring Spawner Abundance (NOSA) are also part of the HLI tables and gives us another independent measure of total spawner abundance (StreamNet, 2021b,

2021c). Hatchery release numbers come from CBR DART who cleaned and compiled the data from Fish Passage Center datasets (Columbia Basin Research, 2026b; Fish Passage Center, n.d.). The StreamNet HLI datasets were composed of data made available from different states' fish and wildlife services as well as tribal communities throughout the Columbia River Basin.

Some other sources were explored, including NPCC hatchery tracker, SDDT, and WDFW fortress websites. Although the sites contained useful data on hatchery releases and adult spawners, there were gaps in data coverage. Overall, CBR DART datasets offered more extensive coverage of the available data than NPCC hatchery trackers did. Since SDDT and WDFW Fortress offered mainly spawner data, StreamNet HLIs and CBR DART datasets remained the primary source of data for this project.

2.2 Data Access and Analysis

This project created reproducible data access pipelines in R to extract and aggregate the juvenile outmigrant, hatchery release, and spawner abundance data. Specifically, three StreamNet HLI tables and the two CBR DART datasets were used. The three HLI tables were accessed through the R package rCAX (Holmes & Williams, 2023) which provided access to NOSA, JuvOut, and RperS. The DART Hatchery Releases data were retrieved using DART's CSV export URLs and cleaned with a single reusable R function that pulls each source for a specific population or ESU/DPS, passing years and subsets as arguments. For both the Streamnet HLI and DART data, custom functions are called once for each subset of data instead of writing duplicate code. Data for both the Upper Columbia River basin and the Snake River basin; and both Chinook salmon and steelhead were included in the data aggregation and analysis.

To connect hatchery release data to juvenile outmigrants and spawner abundance data on the same aggregation level, a crosswalk table that was built using HUC_8 code in NOAA's Watershed Boundary Database (NOAA Fisheries, n.d.) matching NMFS populations was used. Next, the brood year was determined based on the year of outmigration, accounting for the average outmigration time of different salmonids. For example, Snake River Chinook were assigned a brood year that was offset by two years relative to the outmigration year. Meanwhile, for Upper Columbia Chinook and Steelhead, a weighted average formula for calculating brood years due to varying ages of migrating salmonids was used. Using these alignments, a juvenile outmigrant per spawner ratio could be generated for each population and brood year across the four basin/species subsets to begin looking into productivity trends.

3. Results

3.1 Upper Columbia Chinook

Juvenile outmigrant data were reported for the Entiat River, Methow River, and Wenatchee River populations within the Upper Columbia Spring Chinook ESU. At least some data points were available across the entire study period for each population, with Wenatchee River reporting an unbroken series throughout the time frame, whereas Entiat River and Methow River had gaps at the earlier end of the data series (Fig. 1). Total Spawner Abundance Including Jacks (TSAIJ) and Naturally Occurring Spawner Abundance Including Jacks (NOSAIJ) measured by NOSA preceded juvenile outmigrant data by several decades in all three cases. The Naturally Occurring Spawner Abundance Excluding Jacks (NOSAEJ) measurement had inconsistent coverage across the period but tended to disappear almost entirely starting around 1980 (Fig. 4 and Fig. 5).

Natural origin total juvenile outmigrants showed difference between populations (Fig. 2). The largest numbers came out of Wenatchee River with an initial increase followed by a decrease and stabilization at lower numbers. Entiat River and Methow River consistently yielded fewer juvenile outmigrants than Wenatchee River. JuvOut per spawner was also unequal between the three rivers. While the Entiat River saw an overall positive trend in the latest brood years analyzed, Methow River had consistently lower figures (Fig. 3).

Comparing total spawner abundance reported by NOSA and RperS shows similar trends between the data sources in all but a few instances (Fig. 8). For example, there was a significant discrepancy in 2003 in the total number of Entiat River spring Chinook spawners counted by the two services, which matched expectations in terms of measurement methodology and agency source. The most abundant spawners came from Wenatchee River, while Entiat and Methow River saw fewer but more erratic counts of total spawners.

There is extensive data on hatchery releases into the Upper Columbia spring Chinook subset from various hatcheries/sites (Fig. 6 and Fig. 7). Several of these sites, which released more than one million fish in total, had especially large cumulative releases: Leavenworth Hatchery and Leavenworth NFH showed substantial volumes throughout the late 1970s, 1980s, and 2000s before experiencing a drop in numbers later. Other significant sources include Eastbank Hatchery, Chihuahua Hatchery, Ringold Springs Hatchery, and Winthrop Hatchery.

3.2 Upper Columbia Steelhead

Data were collected on total JuvOut for four National Marine Fisheries Service (NMFS) population units in the Upper Columbia Steelhead ESU: Entiat River, Methow River, Okanogan River, and Wenatchee River. While Wenatchee River had data extending back to the earliest dates in the subset, Entiat River, Methow River, and Okanogan River started collecting their data somewhat later in time (Fig. 9). There were breaks in data at the end of the last couple of years for all the populations except Wenatchee River. NOSA datasets exhibited a similar trend where

Wenatchee River had been collecting data earlier compared to Entiat River, Methow River, and Okanogan River (Fig. 12 and Fig. 13).

Total natural JuvOut showed variation between the populations (Fig. 10). Specifically, Wenatchee River had very high peaks of migration midway throughout the dataset, whereas other populations had less pronounced changes over time and had relatively constant numbers. The ratio of JuvOut per spawner differed between populations (Fig. 11). Entiat River, Methow River, and Wenatchee River demonstrated considerable variability within brood years, but Methow River and Wenatchee River ratios increased in recent years.

Entiat River and Methow River had nearly identical trends between NOSA and RperS spawners for each year throughout the datasets (Fig. 16). In the meantime, RperS estimates for Okanogan River and Wenatchee River deviated significantly from the NOSA estimates. More specifically, estimates using the RperS metric tended to be noticeably lower than those based on NOSA in some years within each population. The Methow River population hosted the largest number of spawners, whereas Entiat River had the lowest abundance estimates.

There were several hatcheries involved in the release of spring Chinook salmon for the Upper Columbia Steelhead subset compared to the spring Chinook subset (Fig. 14 and Fig. 15). The largest volumes of releases above one million occurred for the Wells hatchery with a decreasing trend following an early peak. Other hatcheries with considerable volumes above one million included Ringold Springs Hatchery, Winthrop NFH, and Turtle Rock Hatchery. Lower yet steady release amounts took place at the Chelan Hatchery, Chihuahua Hatchery, Eastbank Hatchery, and Leavenworth NFH.

3.3 Snake River Chinook

There are JuvOut data for nineteen NMFS populations in the Snake River spring/summer Chinook ESU. The data availability for JuvOut varied greatly across individual populations but covers nearly the whole period of time recording. While there is coverage for some populations like Tucannon River covering nearly the entire timeframe, other populations, such as North Fork Salmon RIver and Valley Creek have considerable multi year gaps, thus resulting in less time coverage of available data. Data on spawners in the Snake River Lower Mainstem fall run population recorded by NOSA were consistently measured in terms of four metrics: Total Spawner Abundance Including Jacks (TSAIJ), Total Spawner Abundance Excluding Jacks (TSAEJ), Naturally Occurring Spawner Abundance Including Jacks (NOSAIJ), and Naturally Occurring Spawner Abundance Excluding Jacks (NOSAEJ) during almost all of the time period.

There is great variation of natural origin juvenile outmigrant totals across nineteen populations in Snake River (Fig. 18). One of the populations showing maximum values is the

South Fork Salmon River population, whereas the rest of the populations demonstrate relatively low outmigrant totals throughout. Some populations, including Asotin Creek and Pahsimeroi River, are constantly demonstrating low numbers of outmigrant totals. In regard to juvenile outmigrants per spawner (offset = 2 years) values, calculated in accordance with methodology provided in section 2.2, also vary a lot between different populations, yet with most of them demonstrating the same trend and one population, Bear Valley Creek, is distinctly different for at least one brood year.

Spawner abundance values estimated both from NOSA and RperS coincide very well across nearly all nineteen populations considered,values were practically equal for all reporting years except for 1979 when some differences appeared between datasets (Fig. 24). Generally, there is a tendency toward earlier peak value and subsequent decline in spawner abundance, Population size varies dramatically, ranging from constantly small populations such as Camas Creek up to considerably larger populations such as Imnaha River Mainstem.

The list of hatchery releases for Snake River Chinook includes records on ten major hatcheries and many locations where fish is released (Fig. 22 and Fig. 23). The biggest amount of hatchery releases is demonstrated by Rapid River Hatchery and Clearwater Hatchery. In particular, Clearwater Hatchery tends to increase its annual release number throughout all considered reporting years. Dworshak NFH, Sawtooth Hatchery, McCall Hatchery, and Lookingglass Hatchery also play an important role of providing a considerable volume of releases consistently every year. Other relatively small contributors include Kooskia NFH, Pahsimeroi Hatchery, and Nez Perce Tribal Hatchery.

3.4 Snake River Steelhead

JuvOut data was gathered for seventeen NMFS populations belonging to the Snake River Steelhead ESU. Data on JuvOuts are available for each population for nearly the entire duration of time observed. Although the number of observations differs greatly across individual populations, some populations exhibit fairly continuous coverage during observation, including South Fork Salmon River and South Fork Clearwater River populations. At the same time, other populations, such as Secesh River and Lolo Creek populations, feature significant periods of missing values occurring at various points during observations (Fig. 25). The availability of NOSA spawner observations also varies by population. Some populations feature earlier starting dates for observations than others, including Grande Ronde River Upper Mainstream and Joseph Creek populations (Fig. 28 and Fig. 29).

Observations on the overall number of natural origin JuvOuts vary largely across the seventeen populations. While the South Fork Salmon River population demonstrates a sharp increase towards the end of the observed period, the rest of the populations remain almost

unchanged throughout observations (Fig. 26). Calculation of JuvOuts per spawner can be done only for those populations whose brood year offsets are known. These include Clearwater River Lower Mainstem and Little Salmon River populations. Both populations display comparable variability across the observed brood years (Fig. 27).

Total numbers of NOSA spawners and RperS demonstrate high correlation across the majority of populations compared in one graph. Values obtained from both datasets overlap substantially throughout observation (Fig. 32). Some differences in the values occur only for Clearwater River Lower Mainstem and Little Salmon River populations until recently when they started converging again.

Records of hatchery releases feature a total of twelve hatcheries involved in replenishing the Snake River Steelhead population. Among these are Dworshak NFH who has been releasing a maximum number of fishes with the amount decreasing to stabilize at lower levels in the most recent decades, as well as Irrigon Hatchery, Niagara Springs, Magic Valley Hatchery, and Lyons Ferry hatcher with fairly constant release amounts throughout observation (Fig. 30 and Fig. 31). Smaller hatchery release volumes include Big Canyon Hatchery, Hagerman NFH, and Hayden Creek Hatchery releases.

4. Next Steps

We successfully created an automated reproducible pipeline to access juvenile abundance and spawner abundance from StreamNet and CBR DART at a subbasin level. However, one remaining challenge is differing levels of aggregation for the hatchery release locations and NMFS population groups. In order to resolve this issue and link hatchery release records to specific NMFS populations, we will either need to manually review each release site to assign a corresponding population group, or aggregate population groups up to the level of HUC8 codes to correspond with hatchery releases.

For the next stage of work, it's intended to construct a SQL database to host and analyze aggregated data. It would ensure consistent maintenance and sharing of the resulting dataset as new data arrives for four basin/species combinations. Furthermore, there are plans to develop statistical models evaluating density dependence impacts and environmental factors affecting juvenile outmigrants per spawner ratios.

References

Columbia Basin Research. (2026a). *DART adult passage counts annual summary for all species* [Data set]. University of Washington. https://cbr.washington.edu/dart/query/adult_annual_sum

Columbia Basin Research. (2026b). *DART hatchery releases with distribution summaries* [Data set]. University of Washington. https://cbr.washington.edu/dart/query/hatch

Ford, M. J., Lindley, S. T., Barnas, K. A., Shelton, A. O., Spence, B. C., Weitkamp, L. A., Holzer, D. M., Boughton, D. A., Holmes, E. E., Myers, J. M., Jordan, C. E., Fish, H., Liermann, M., O'Farrell, M. R., Mantua, N. J., Johnson, R. C., Satterthwaite, W. H., & Williams, T. H. (2025). Abundance trends of Pacific salmon during a quarter century of ESA protection. *Fish and Fisheries, 26*(6), 1087–1106. https://doi.org/10.1111/faf.70019

Holmes, E. E., & Williams, M. (2023). rCAX: Coordinated assessments REST API R client (R package version 1.0.3) [Computer software]. NOAA Fisheries, Northwest Fisheries Science Center, Math Bio Program. https://doi.org/10.5281/zenodo.7402463

Jaeger, W. K., & Scheuerell, M. D. (2023). Return(s) on investment: Restoration spending in the Columbia River Basin and increased abundance of salmon and steelhead. *PLOS ONE, 18*(7), Article e0289246. https://doi.org/10.1371/journal.pone.0289246

Kostow, K. (2012). Strategies for reducing the ecological risks of hatchery programs: Case studies from the Pacific Northwest. *Environmental Biology of Fishes, 94*(2), 285–310. https://doi.org/10.1007/s10641-011-9868-1

Northwest Power and Conservation Council. (2026). *Program tracker: Hatcheries* [Data set]. https://projects.nwcouncil.org/programtracker/modules/data/hatcheries

Pacific States Marine Fisheries Commission. (2026). *SDDT dataset details* [Data set]. https://sddt.psmfc.org/search/datasetdetails/50

StreamNet. (2021a, June 10). *CAP fish HLIs query: JuvOut* [Data set]. https://www.streamnet.org/data/hli/?index=1&perpage=100&hli=JuvOut

StreamNet. (2021b, June 10). *CAP fish HLIs query: NOSA* [Data set]. https://www.streamnet.org/data/hli/?index=1&perpage=100&hli=NOSA

StreamNet. (2021c, June 10). *CAP fish HLIs query: RperS* [Data set]. https://www.streamnet.org/data/hli/?index=1&perpage=100&hli=RperS

U.S. Geological Survey. (n.d.). *Watershed Boundary Dataset (WBD)* [Data set]. U.S. Department of the Interior. https://www.usgs.gov/national-hydrography/watershed-boundary-dataset

Washington Department of Fish and Wildlife. (2026). *Chinook: General information page* [Data set]. https://fortress.wa.gov/dfw/score/score/species/chinook.jsp?species=Chinook

Zabel, R. W., Scheuerell, M. D., McClure, M. M., & Williams, J. G. (2006). The interplay between climate variability and density dependence in the population viability of Chinook salmon. *Conservation Biology, 20*(1), 190–200. https://doi.org/10.1111/j.1523-1739.2006.00300.x

Upper Columbia Chinook Figures

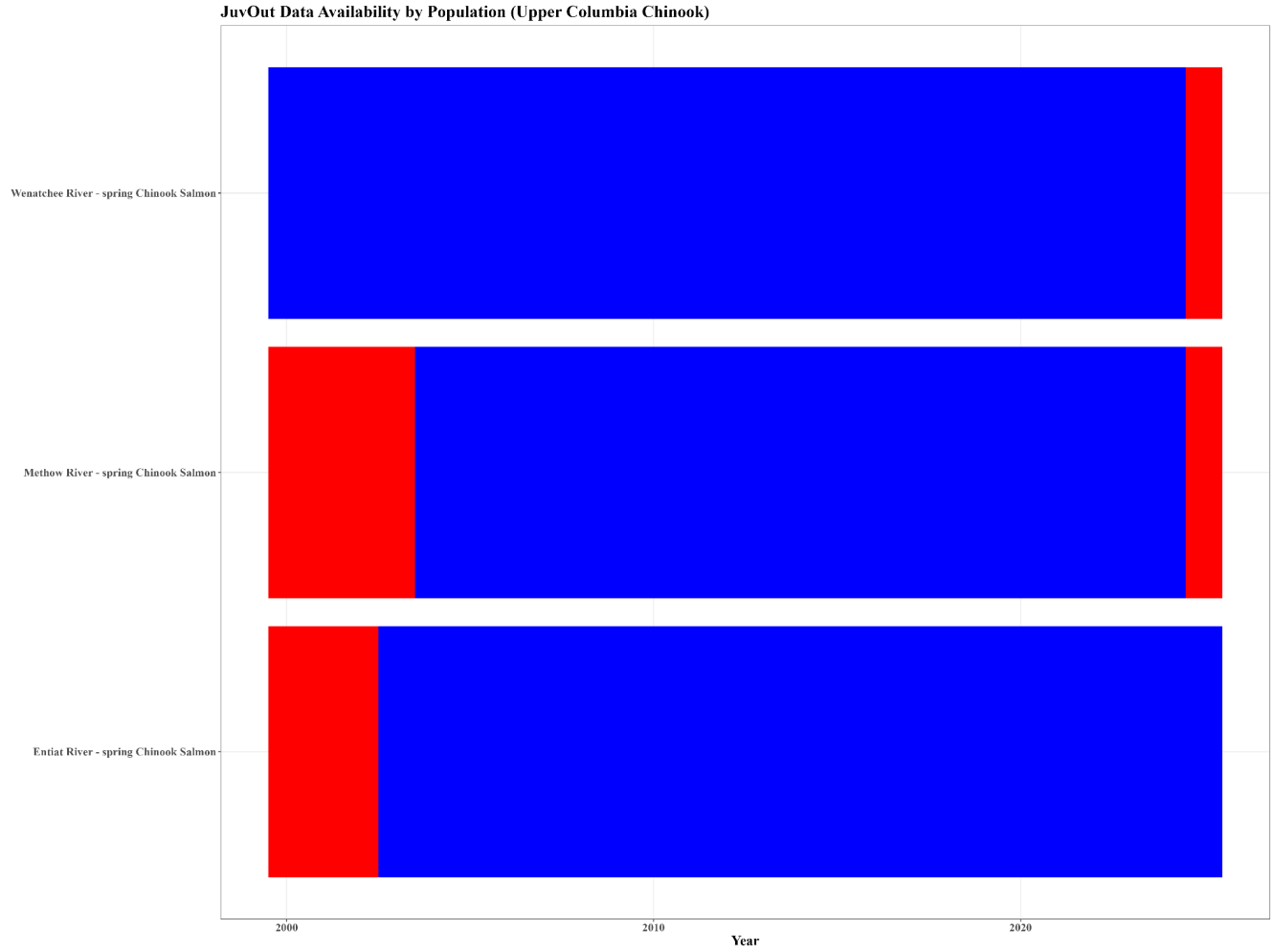


Figure 1: Bar chart showing data availability for StreamNet JuvOut records by population in the Upper Columbia spring Chinook subset (Entiat River, Methow River, and Wenatchee River). Bar segments span through the years with available data for each population,

with blue segments indicating years with available data and red segments indicating years with no data.

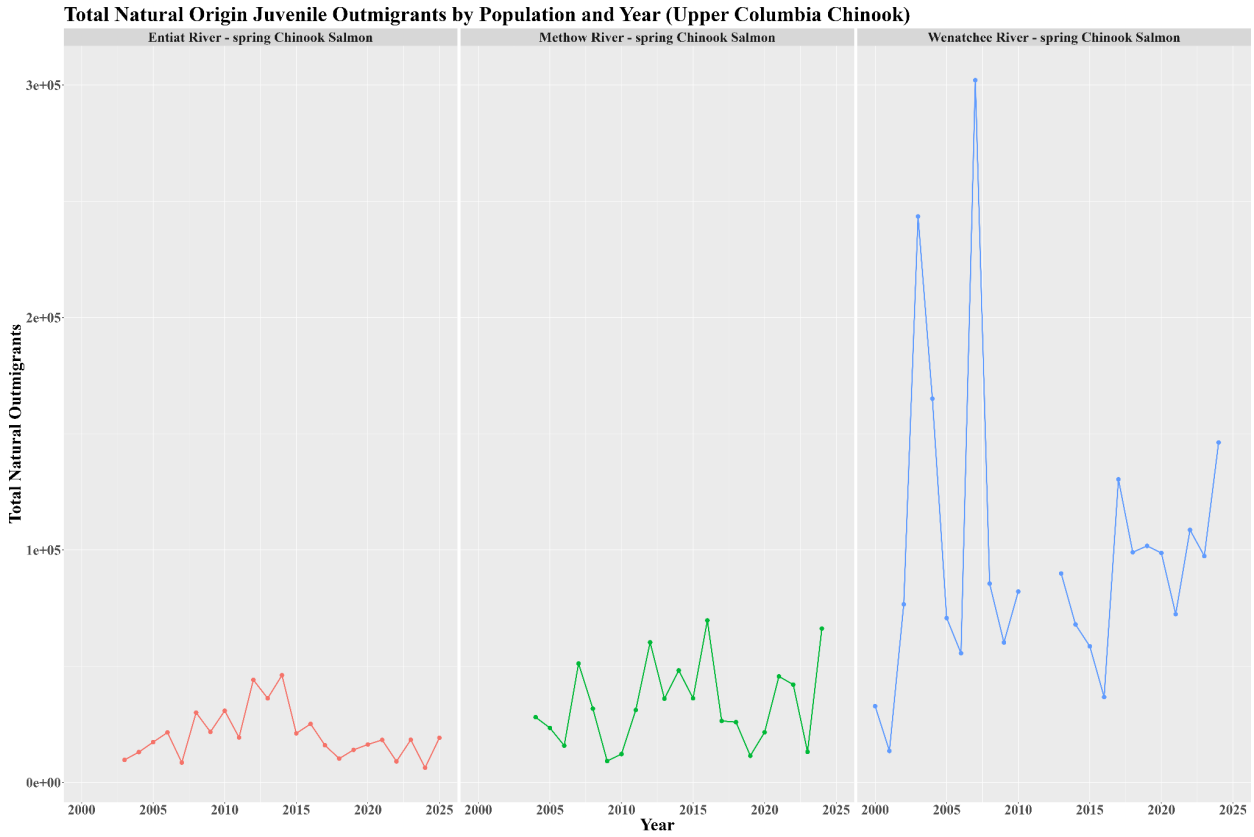


Figure 2: Line graph showing total natural origin juvenile outmigrants by year for each population in the Upper Columbia spring Chinook subset. Populations are distinguished by color (Entiat River in red, Methow River in green, Wenatchee River in blue), with points marking individual annual estimates.

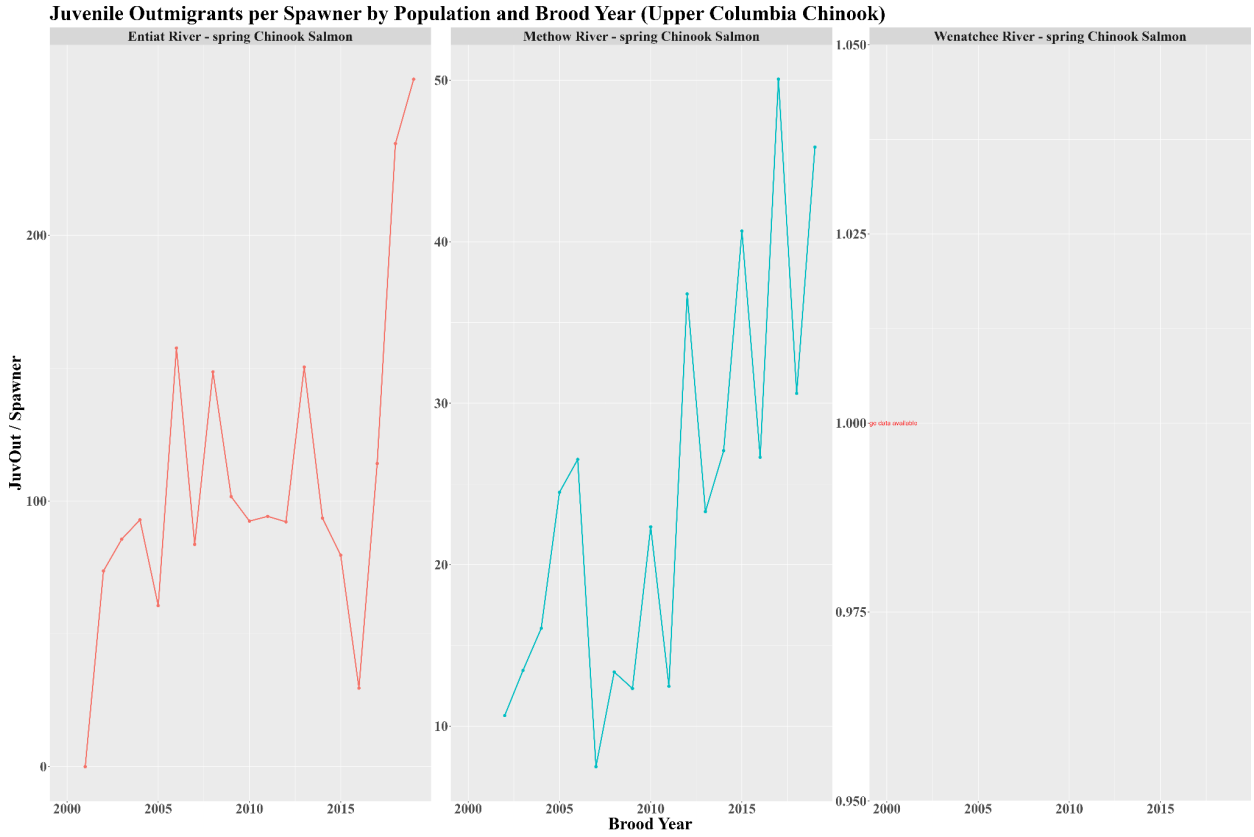


Figure 3: Line graph showing juvenile outmigrants per spawner by brood year for Upper Columbia spring Chinook populations faceted by population (Entiat River and Methow River). Brood year is assigned using the age weighted brood year function described in Methods section 2.2.

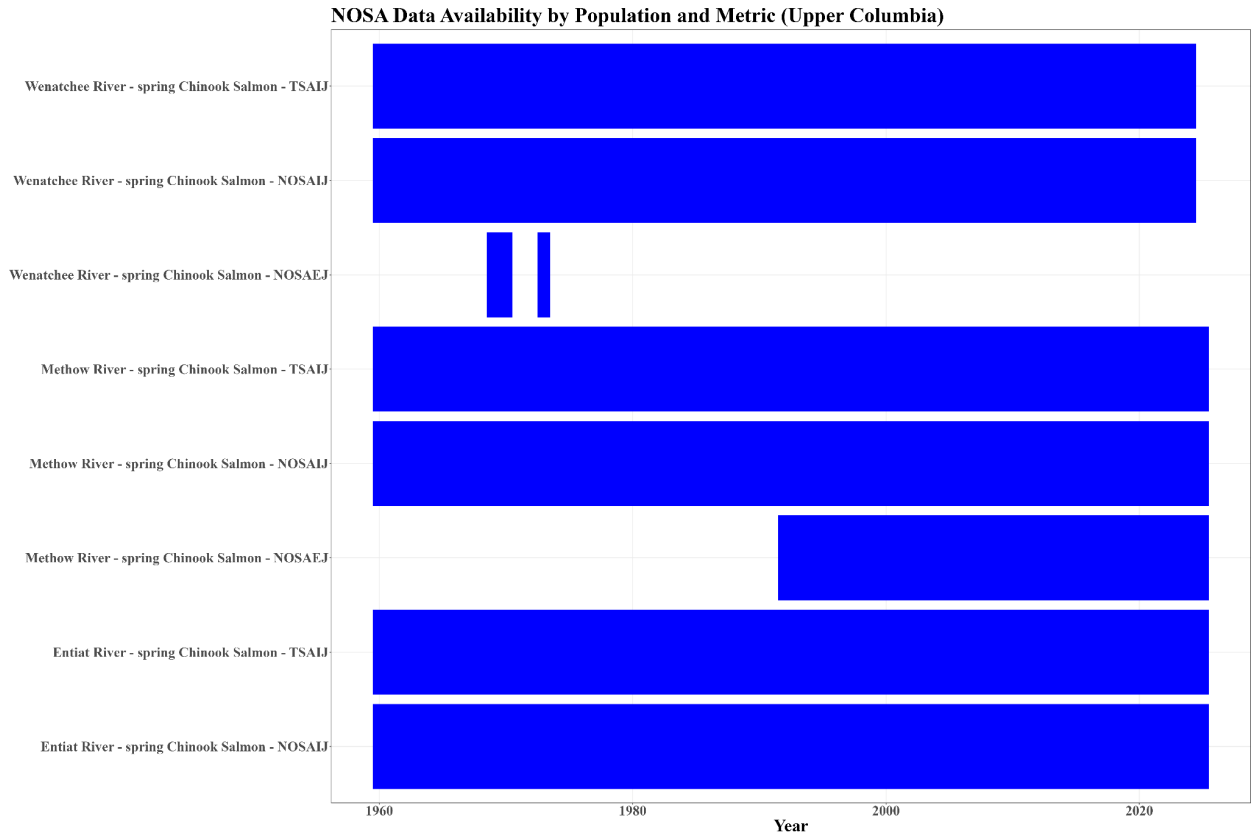


Figure 4: Bar graph showing data availability for StreamNet NOSA records by population and metric for Upper Columbia spring Chinook, including total spawner abundance including jacks (TSAIJ), natural origin spawner abundance including jacks (NOSAIJ) and natural origin spawner abundance excluding jacks (NOSAEJ).

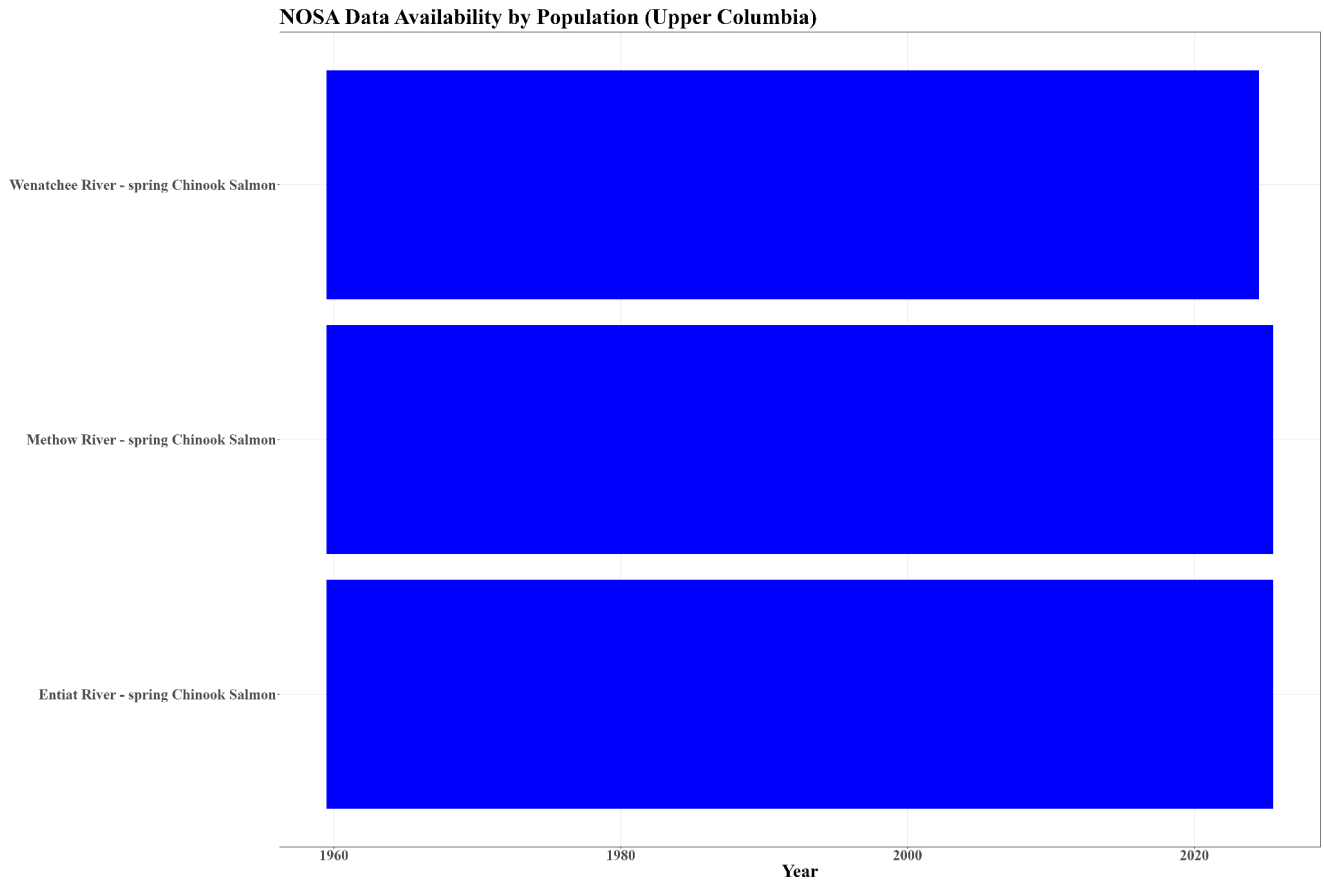


Figure 5: A bar graph showing data availability for StreamNet NOSA records by population for Upper Columbia spring Chinook faceted by population (Entiat River, Methow River, and Wenatchee River).

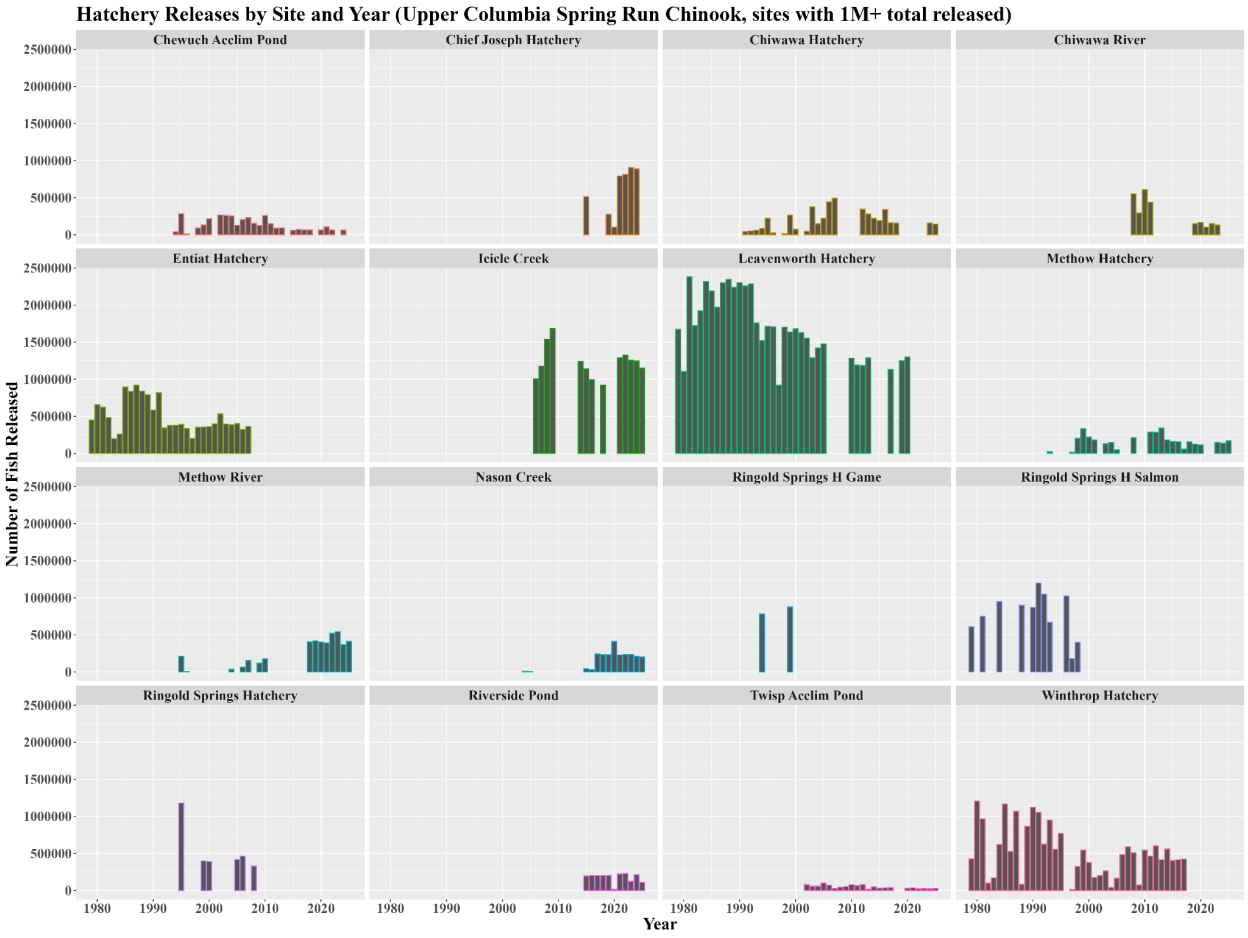


Figure 6: A bar graph showing the number of hatchery-origin fish released by year for each release site in the Upper Columbia spring run Chinook subset, faceted by site. Only sites with more than 1 million total fish released across all years are shown.

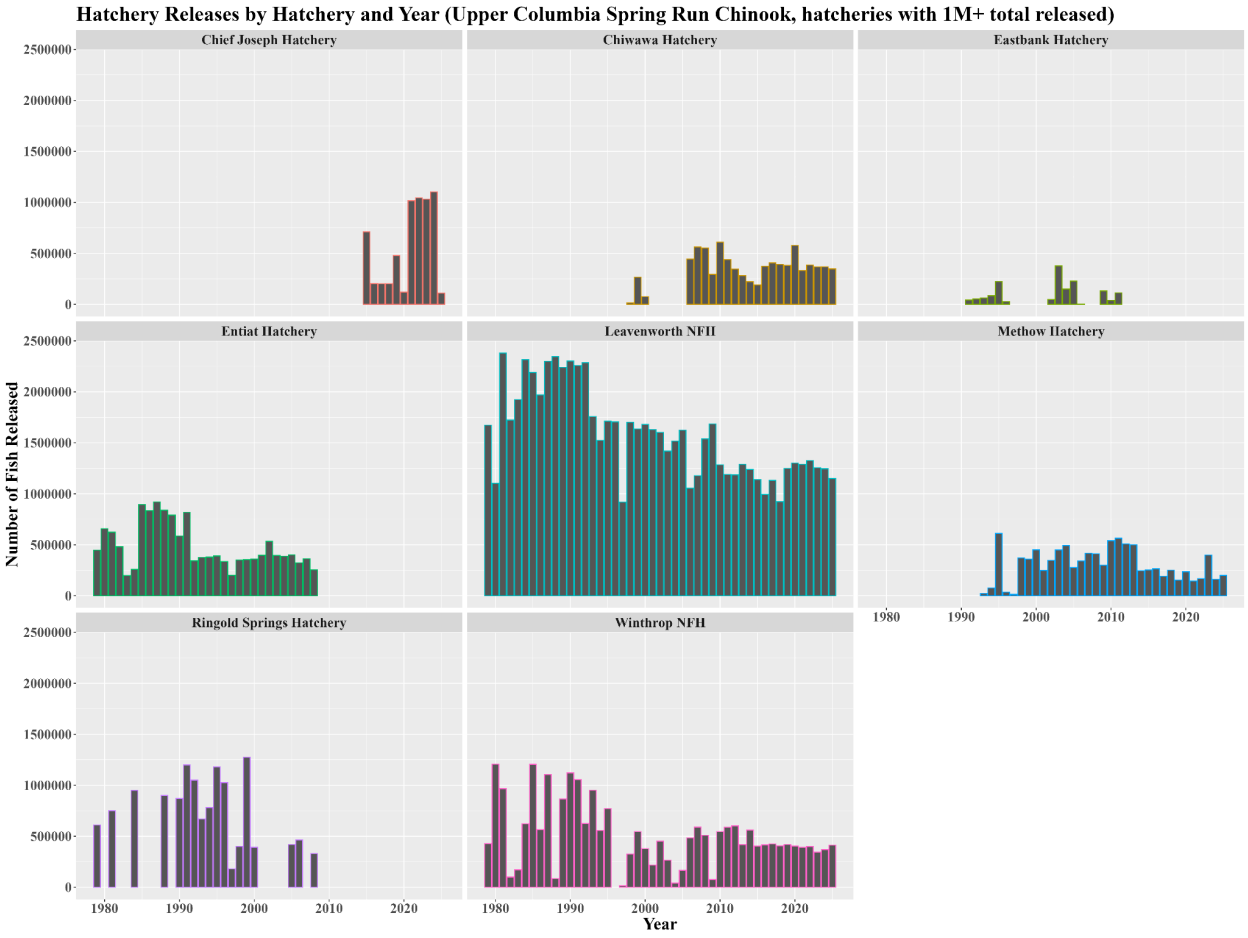


Figure 7: A bar graph showing the number of hatchery origin fish released by year for each hatchery in the Upper Columbia spring run Chinook subset, faceted by hatchery. Only hatcheries with more than 1 million total fish released across all years are shown.

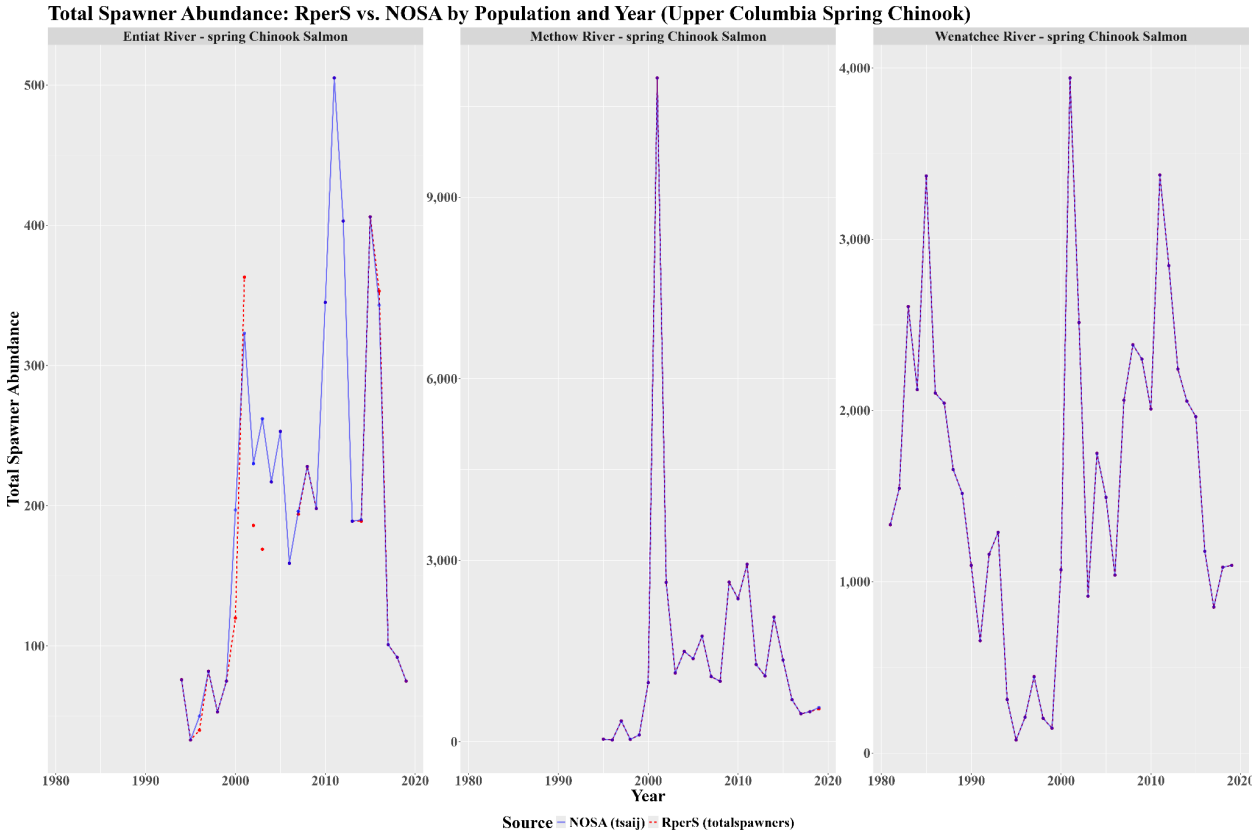


Figure 8: A line graph showing total spawner abundance by year for Upper COlumbia spring Chinook populations, faceted by population (Entiat River, Methow River, and Wenatchee River), comparing estimates from NOSA (blue solid line) and RperS (red dashed line) to show agreement and divergence in numbers between the two data sources.

Upper Columbia Steelhead Figures

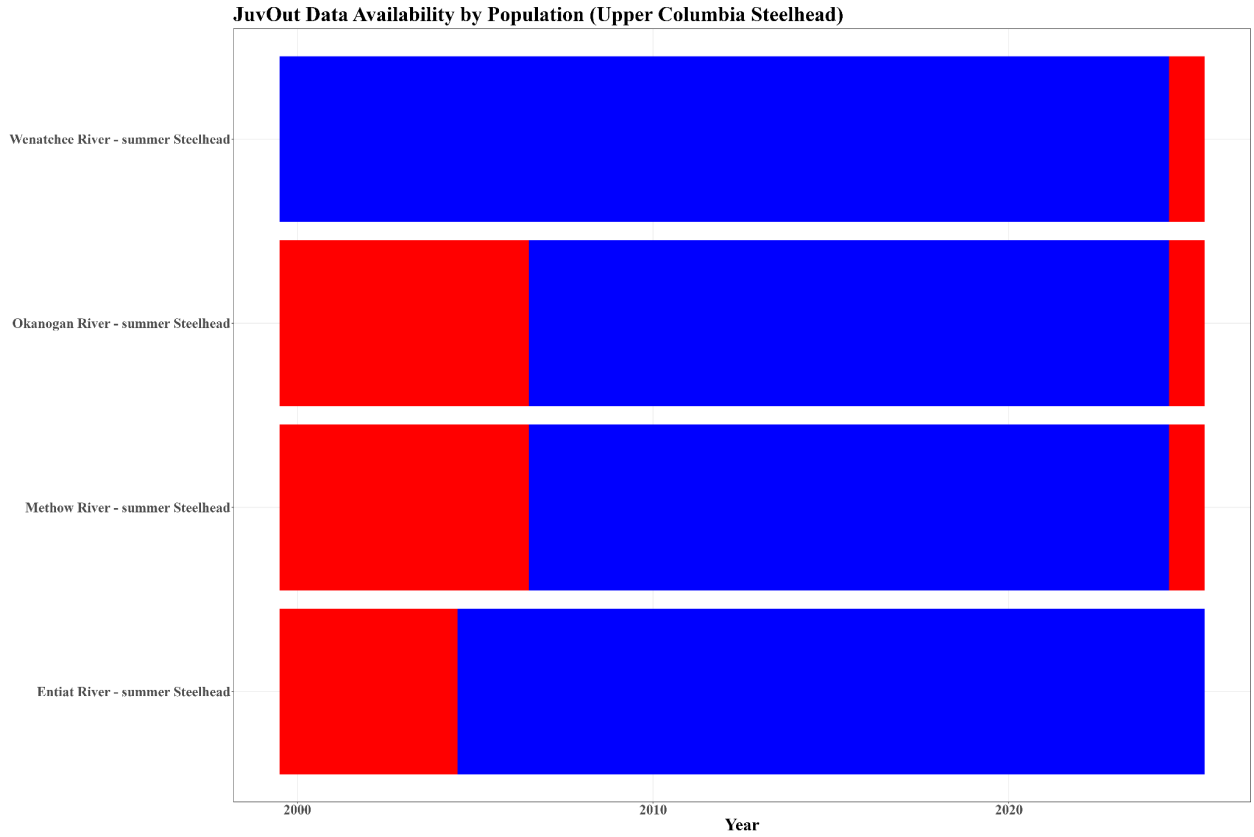


Figure 9: A bar graph showing data availability for StreamNet JuvOut records by population in the Upper Columbia Steelhead subset (Entiat River, Methow River, Okanogan River, and Wenatchee River). Blue segments indicate years with available data and red segments indicate years with no data.

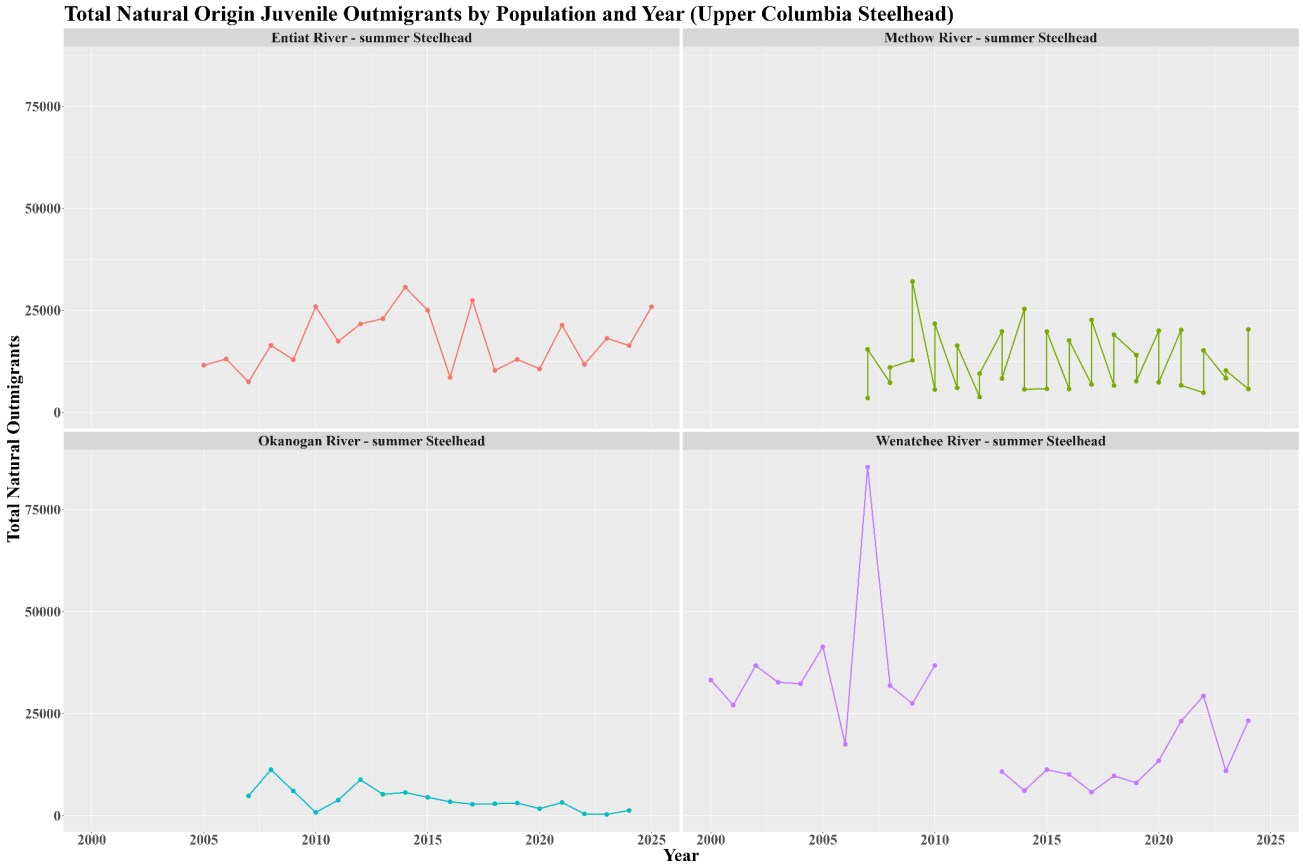


Figure 10: A line graph showing total natural origin juvenile outmigrants by year for each population in the Upper Columbia Steelhead subset, faceted by population (Entiat River, Methow River, Okanogan River, and Wenatchee River).

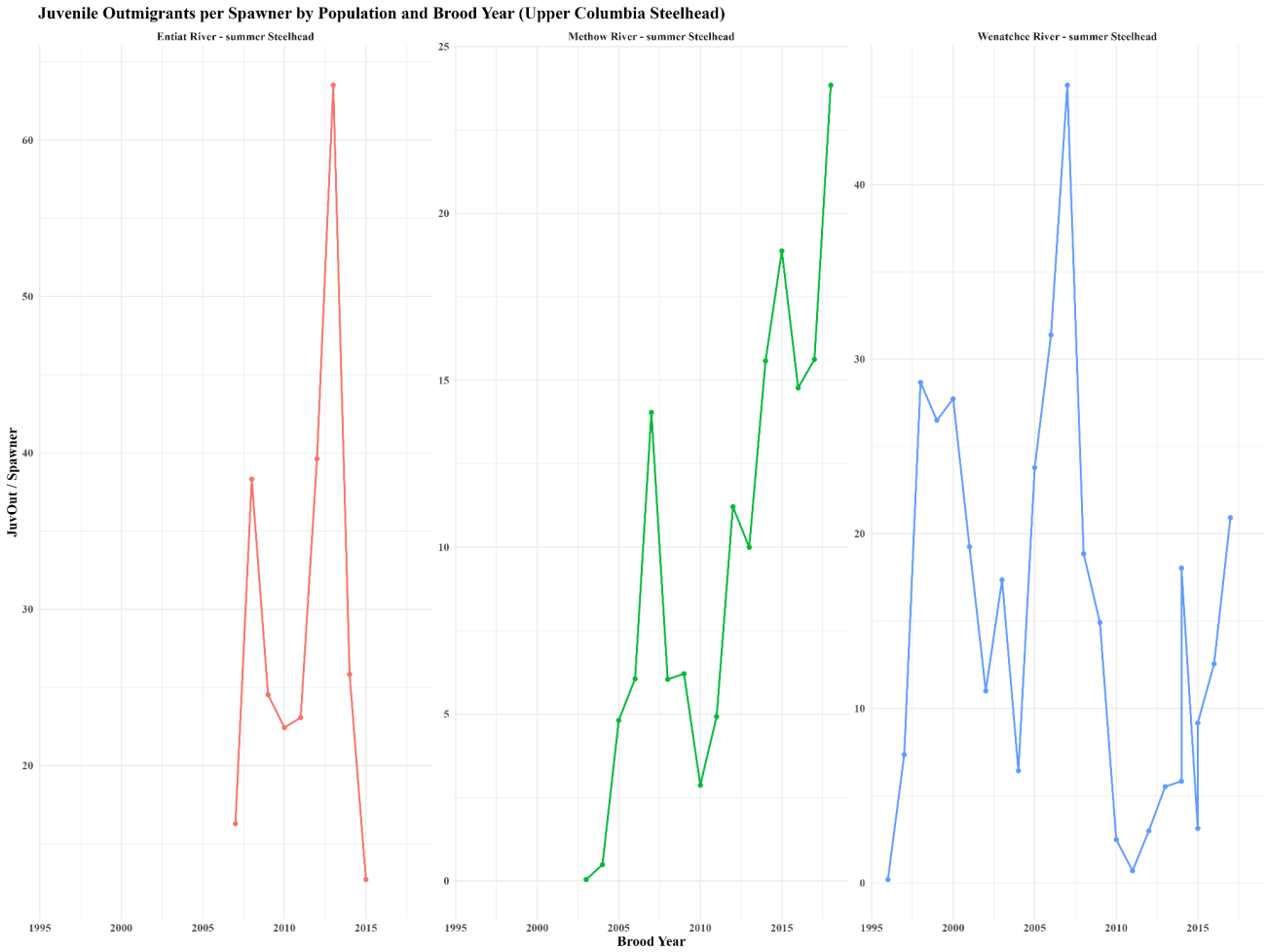


Figure 11: A line graph showing juvenile outmigrants per spawner by brood year for Upper Columbia Steelhead populations, faceted by population (Entiat River, Methow River, and Wenatchee River). Brood year is assigned using the age weighted brood year function described in Methods section 2.2.

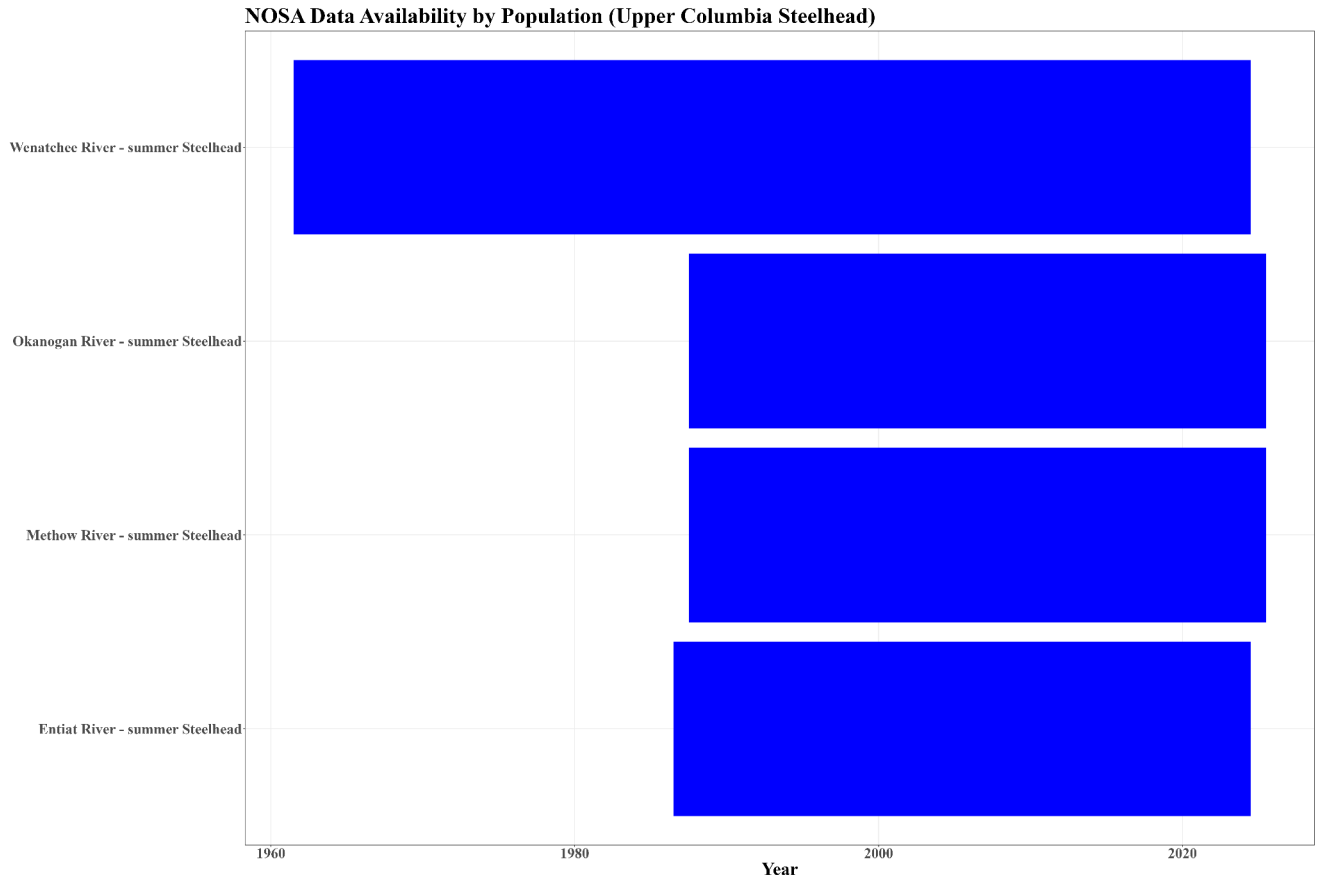


Figure 12: A bar graph showing data availability for StreamNet NOSA records by population for Upper Columbia Steelhead.

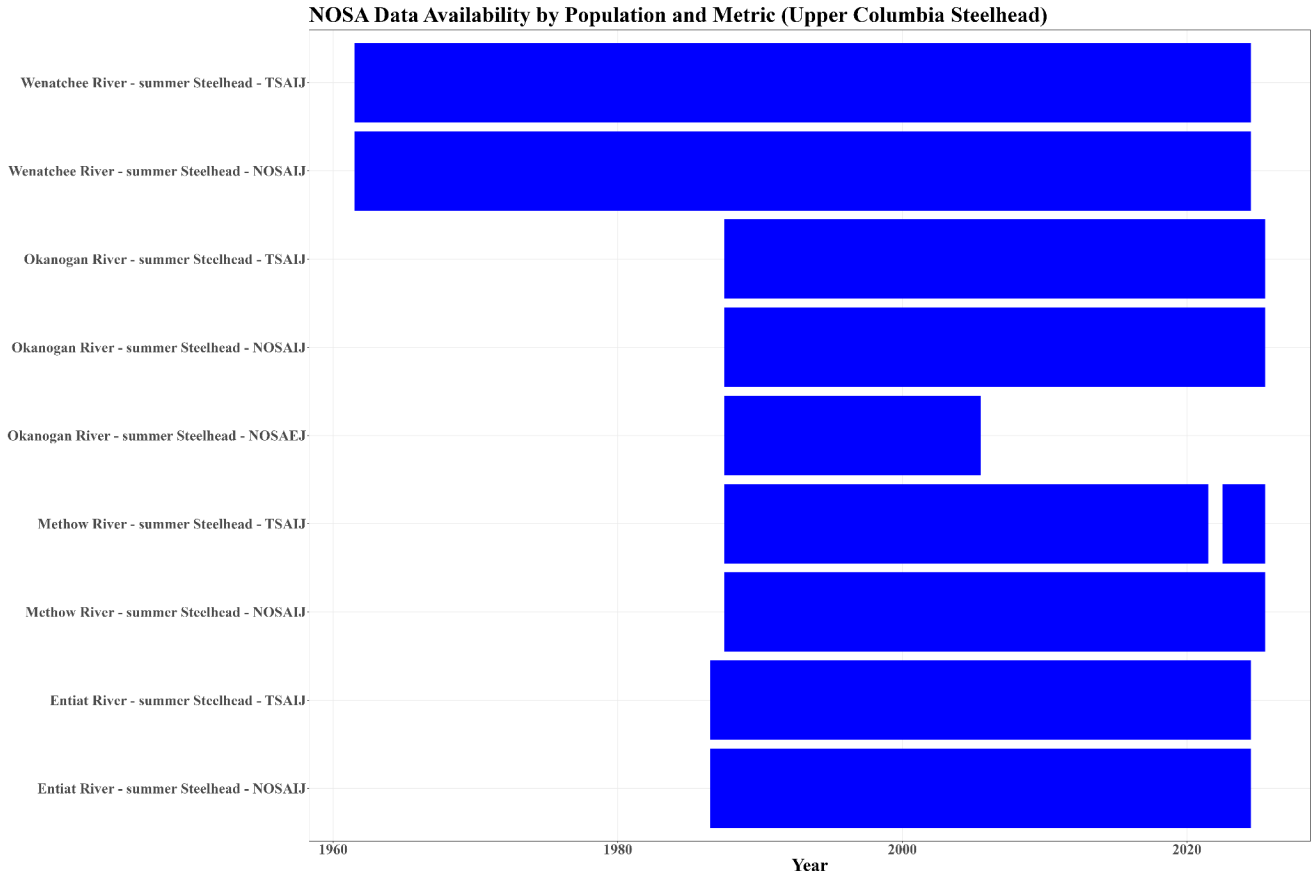


Figure 13: A bar graph showing data availability for StreamNet NOSA records by population and metric for Upper Columbia Steelhead, including total spawner abundance including jacks (TSAIJ), naturally occurring spawner abundance including jacks (NOSAIJ), and naturally occurring spawner abundance excluding jacks (NOSAEJ).

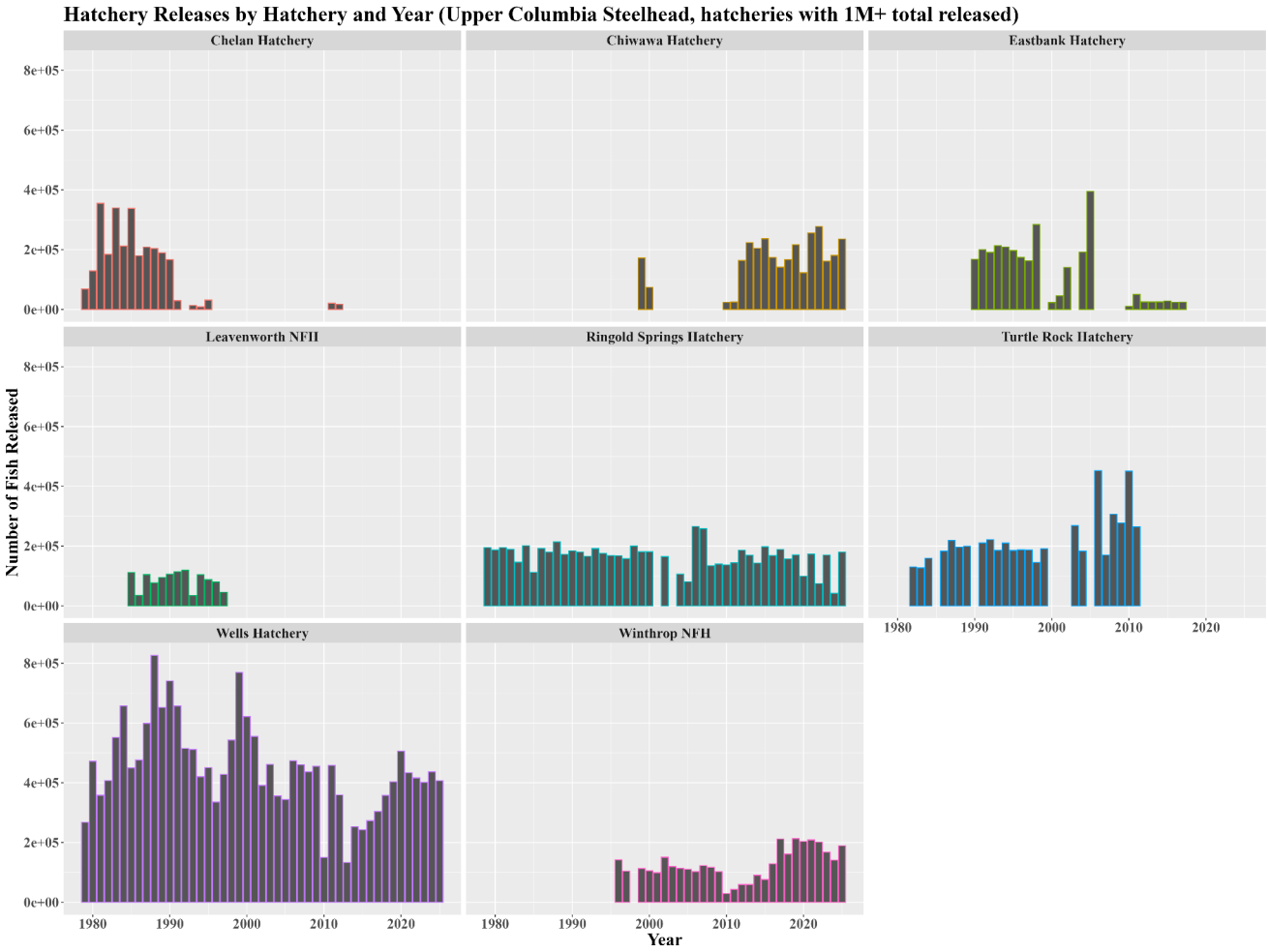


Figure 14: A bar graph showing the number of hatchery origin fish released by year for each hatchery in the Upper Columbia Steelhead subset, faceted by hatchery. Only hatcheries with more than 1 million total fish released across all years are shown.

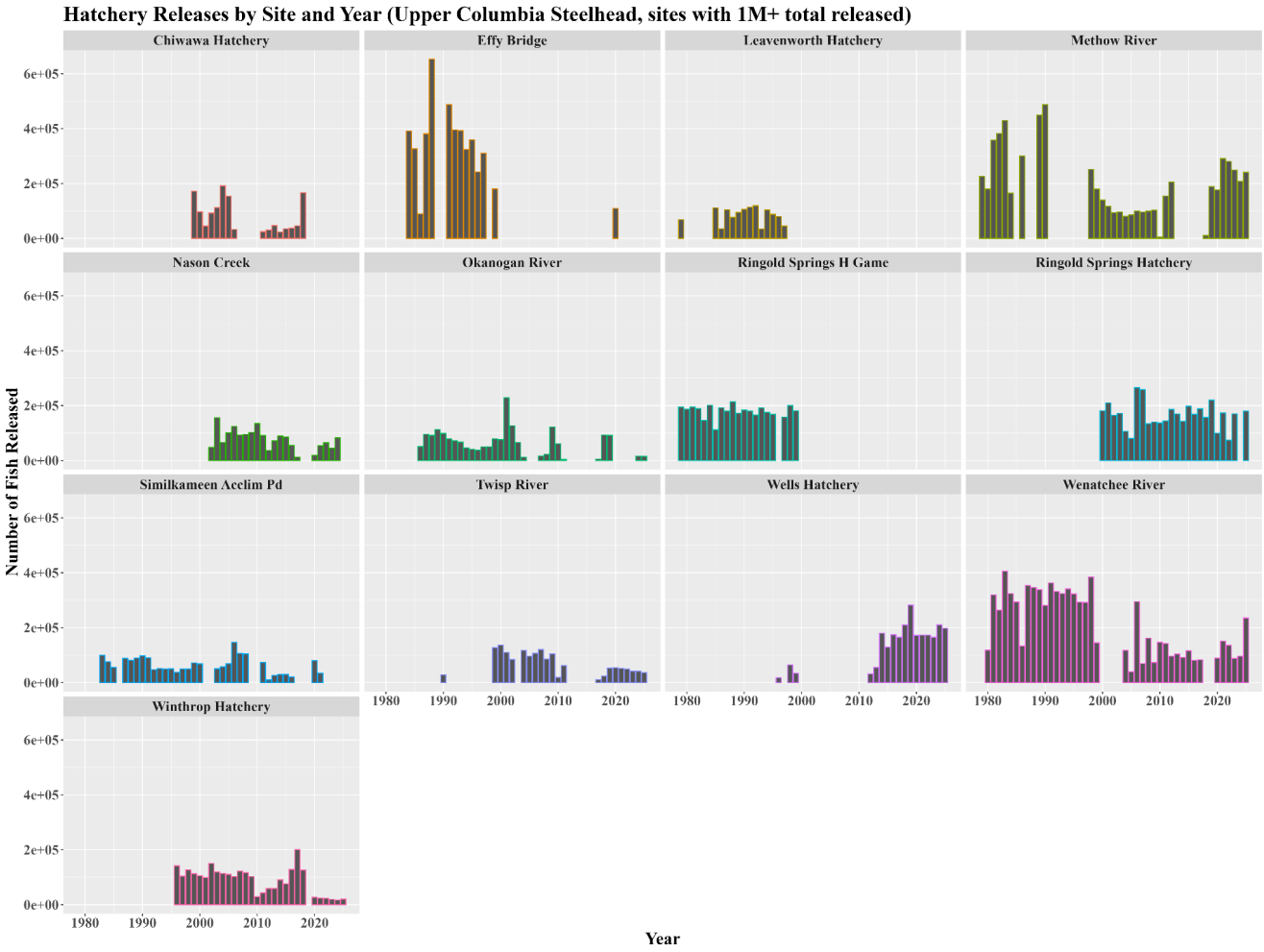


Figure 15: A bar graph showing the number of hatchery origin fish released by year for each release site in the Upper Columbia Steelhead subset, faceted by site. Only sites with more than 1 million total fish released across all years are shown.

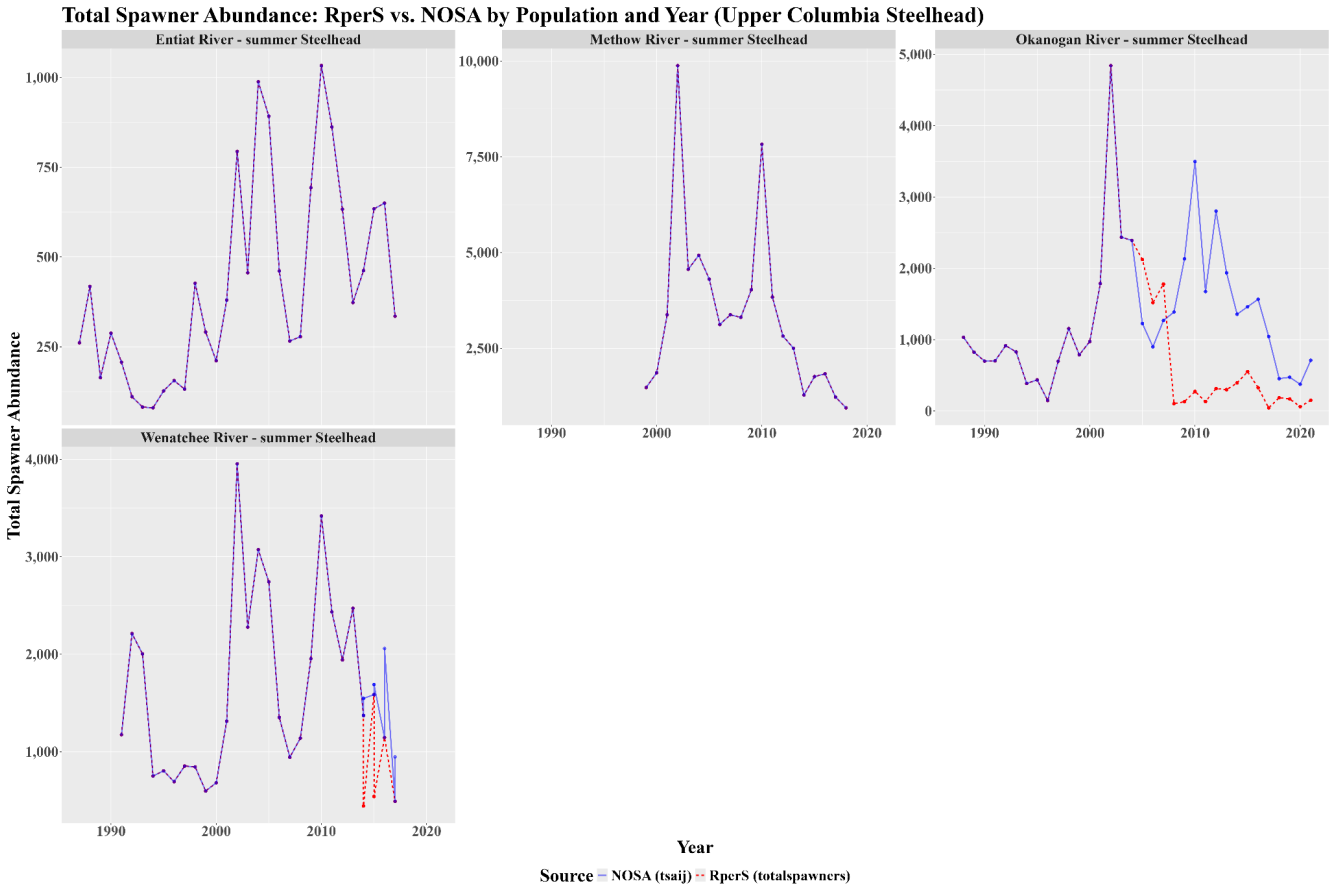


Figure 16: A line graph showing total spawner abundance by year for Upper Columbia Steelhead populations, faceted by population, comparing estimates from NOSA (blue solid line) and RperS (red dashed line).

Snake River Chinook Figures

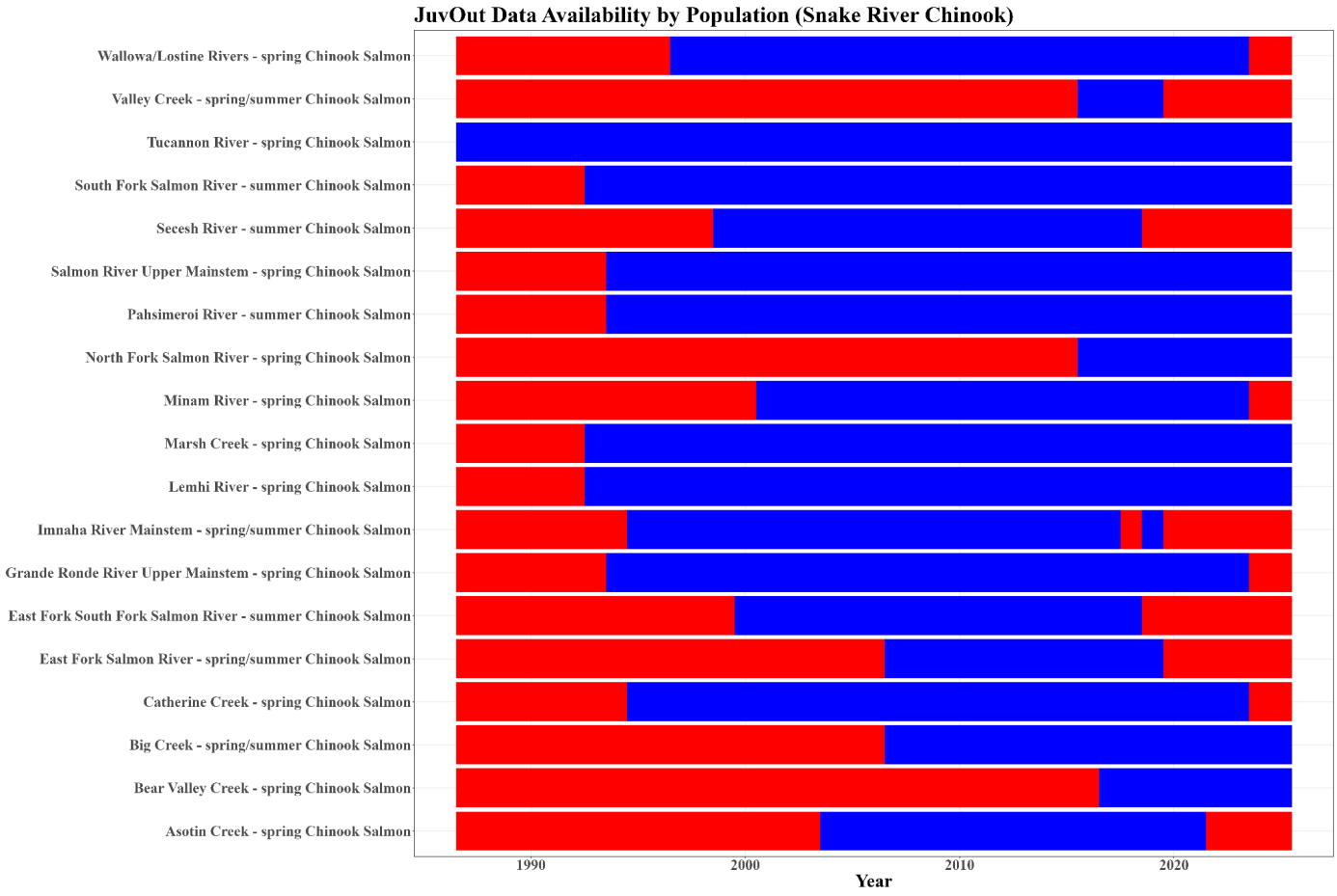


Figure 17: A bar graph showing data availability for StreamNet JuvOut records by population in the Snake River spring/summer Chinook subset. Blue segments indicate years with available data and red segments indicate years with no data.

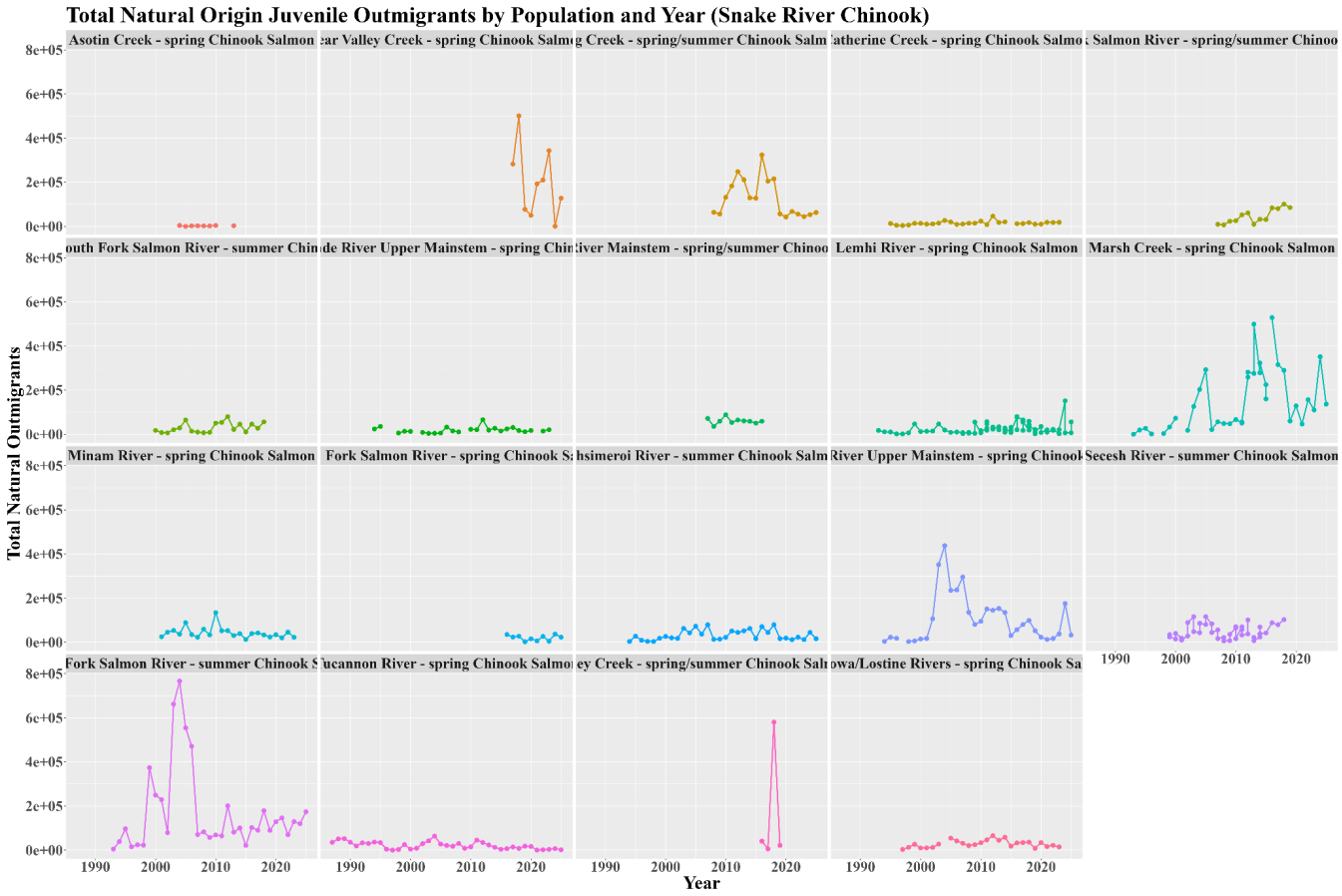


Figure 18: A line graph showing total natural origin juvenile outmigrants by year for each population in the Snake River spring/summer Chinook subset, faceted by population.

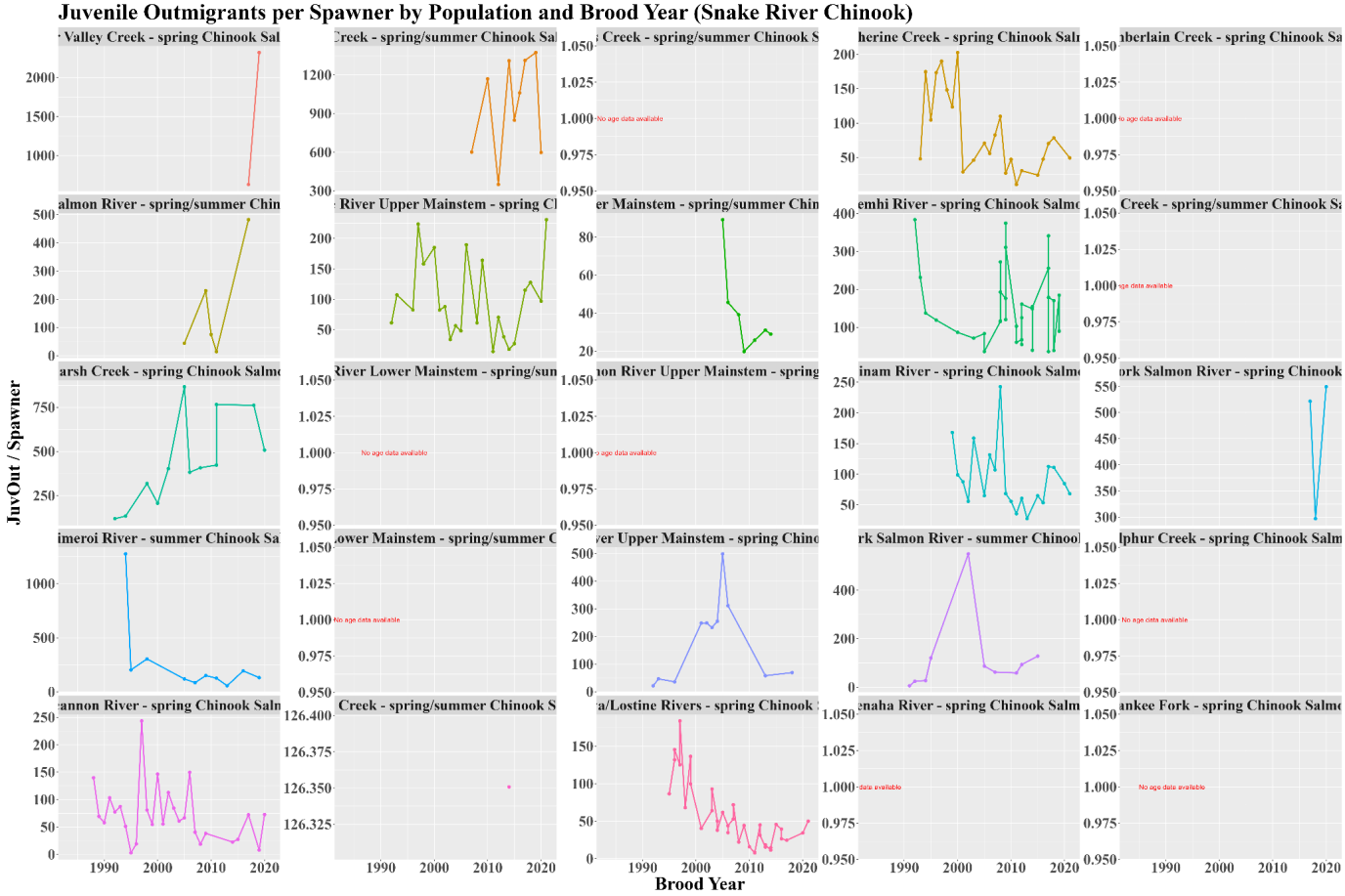


Figure 19: A line graph showing juvenile outmigrants per spawner by brood year for Snake River spring/summer Chinook populations, faceted by population. Brood year is assigned using a fixed two year offset from outmigration year as described in Methods section 2.2.

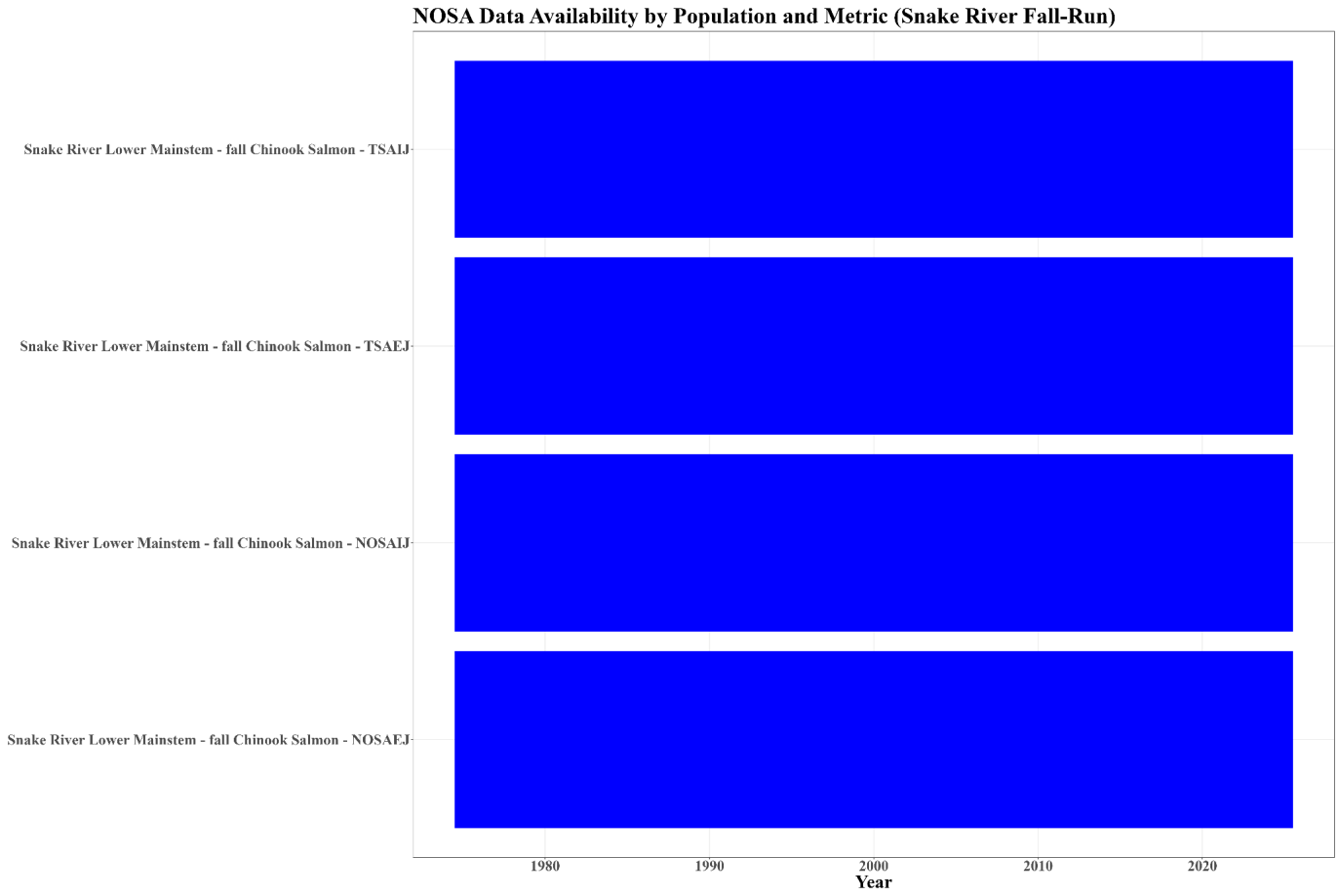


Figure 20: A bar graph showing data availability for StreamNet NOSA records by population and metric for Snake River fall-run Chinook. Metrics include total spawner abundance including jacks (TSAIJ), total spawner abundance excluding jacks (TSAEJ), naturally occurring spawner abundance including jacks (NOSAIJ), and naturally occurring spawner abundance excluding jacks (NOSAEJ).

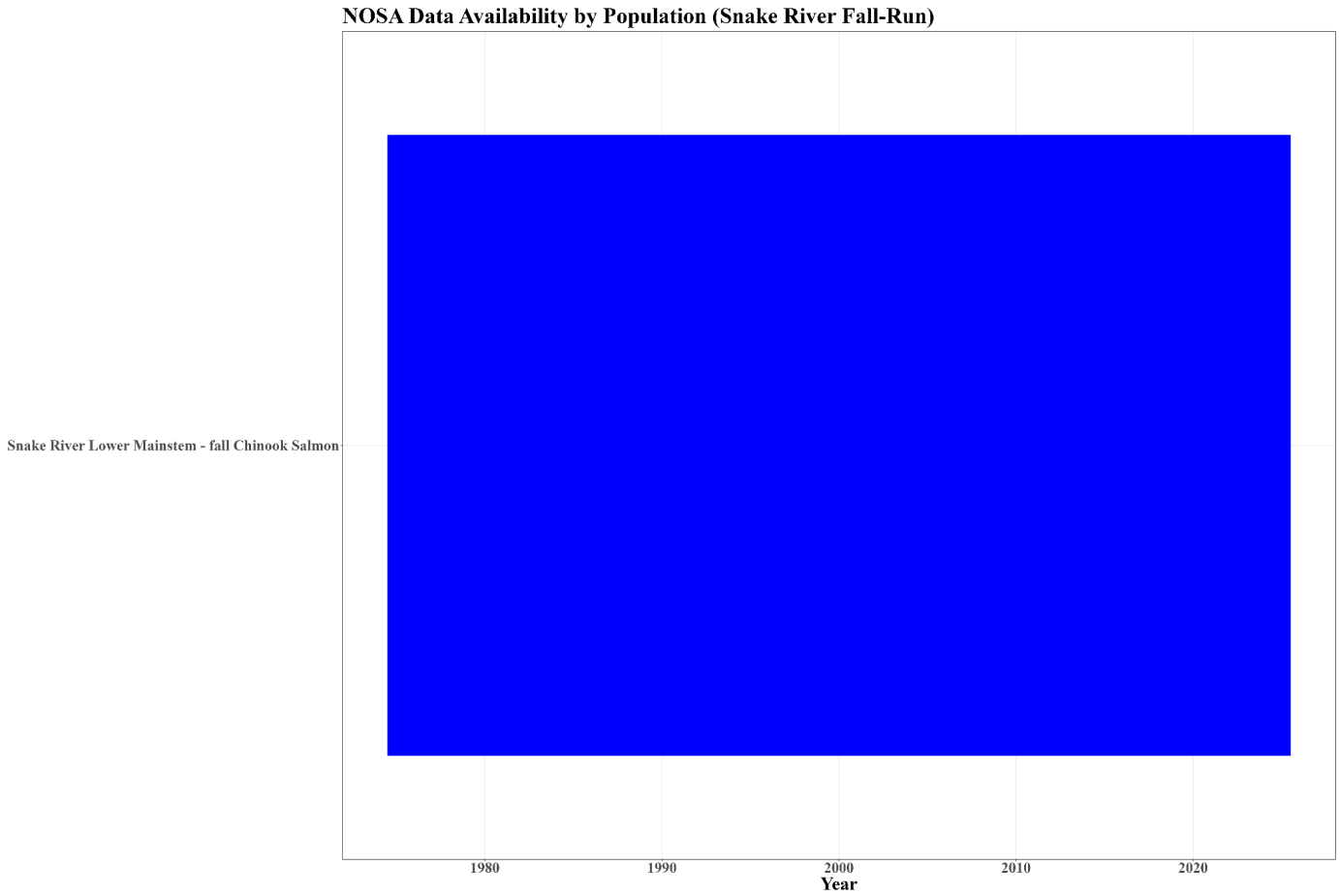


Figure 21: A bar graph showing data availability for StreamNet NOSA records by population for Snake River fall-run Chinook.

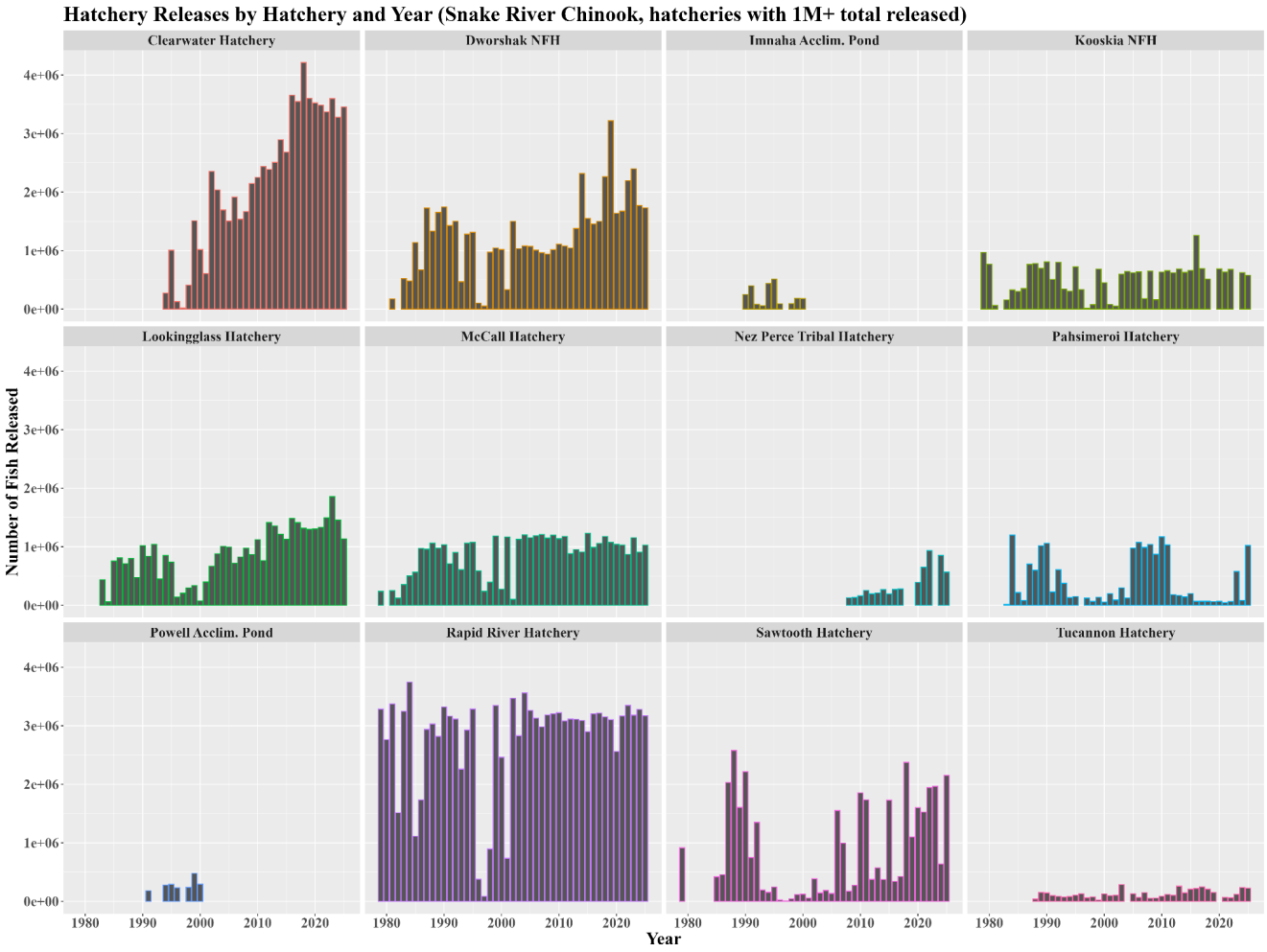


Figure 22: A bar graph showing the number of hatchery origin fish released by year for each hatchery in the Snake River Chinook subset, faceted by hatchery. Only hatcheries with more than 1 million total fish released across all years are shown.



Figure 23: A bar graph showing the number of hatchery origin fish released by year for each release site in the Snake River Chinook subset, facet by site. Only sites with more than 1 million total fish released across all years shown.

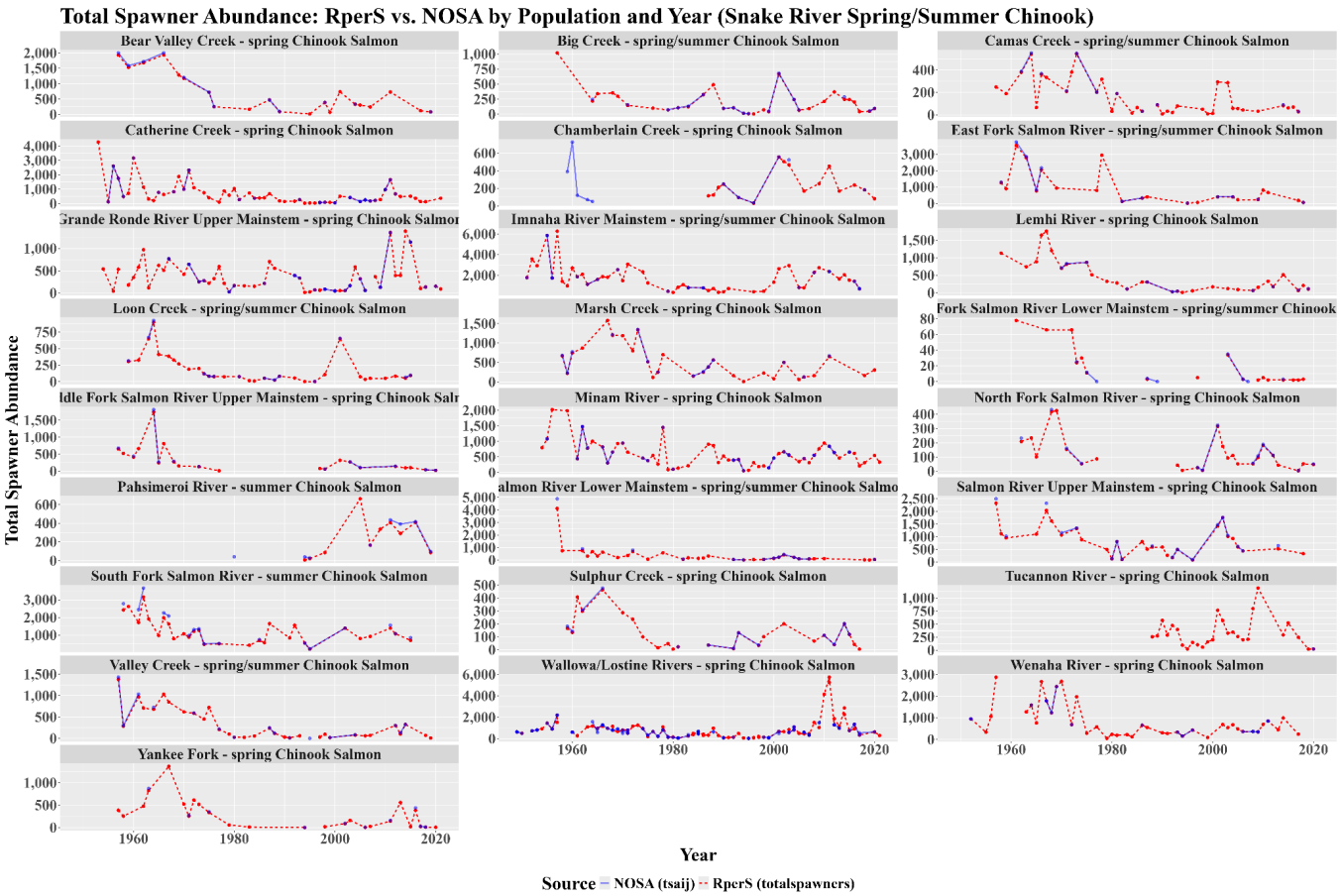


Figure 24: A line graph showing total spawner abundance by year for Snake River spring/summer Chinook populations, faceted by population, comparing estimates from NOSA (blue solid line) and RperS (red dashed line).

Snake River Steelhead Figures

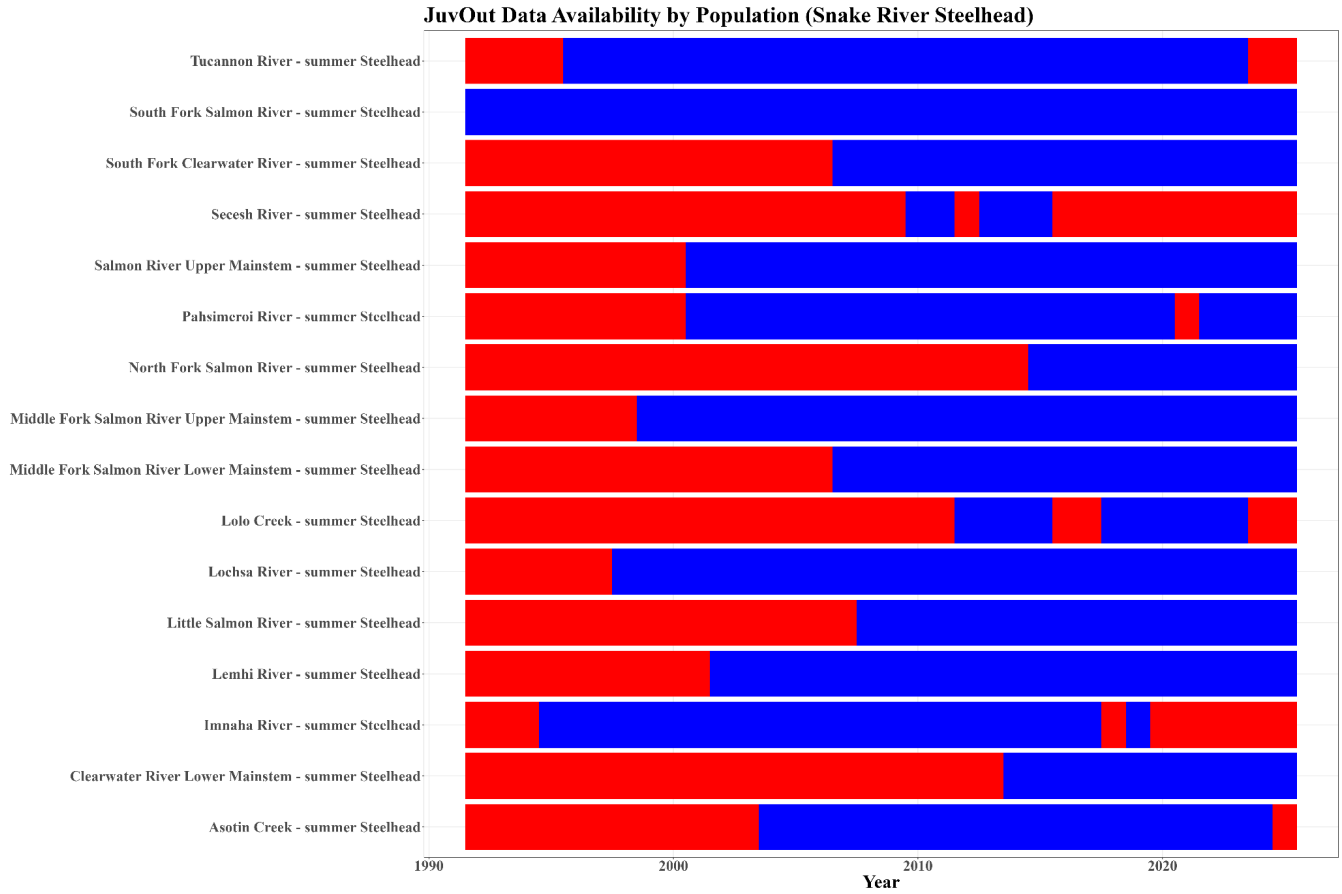


Figure 25: A bar graph showing data availability for StreamNet JuvOut records by population in the Snake River Steelhead subset. Blue segments indicate years with available data and red segments indicate years with no data.

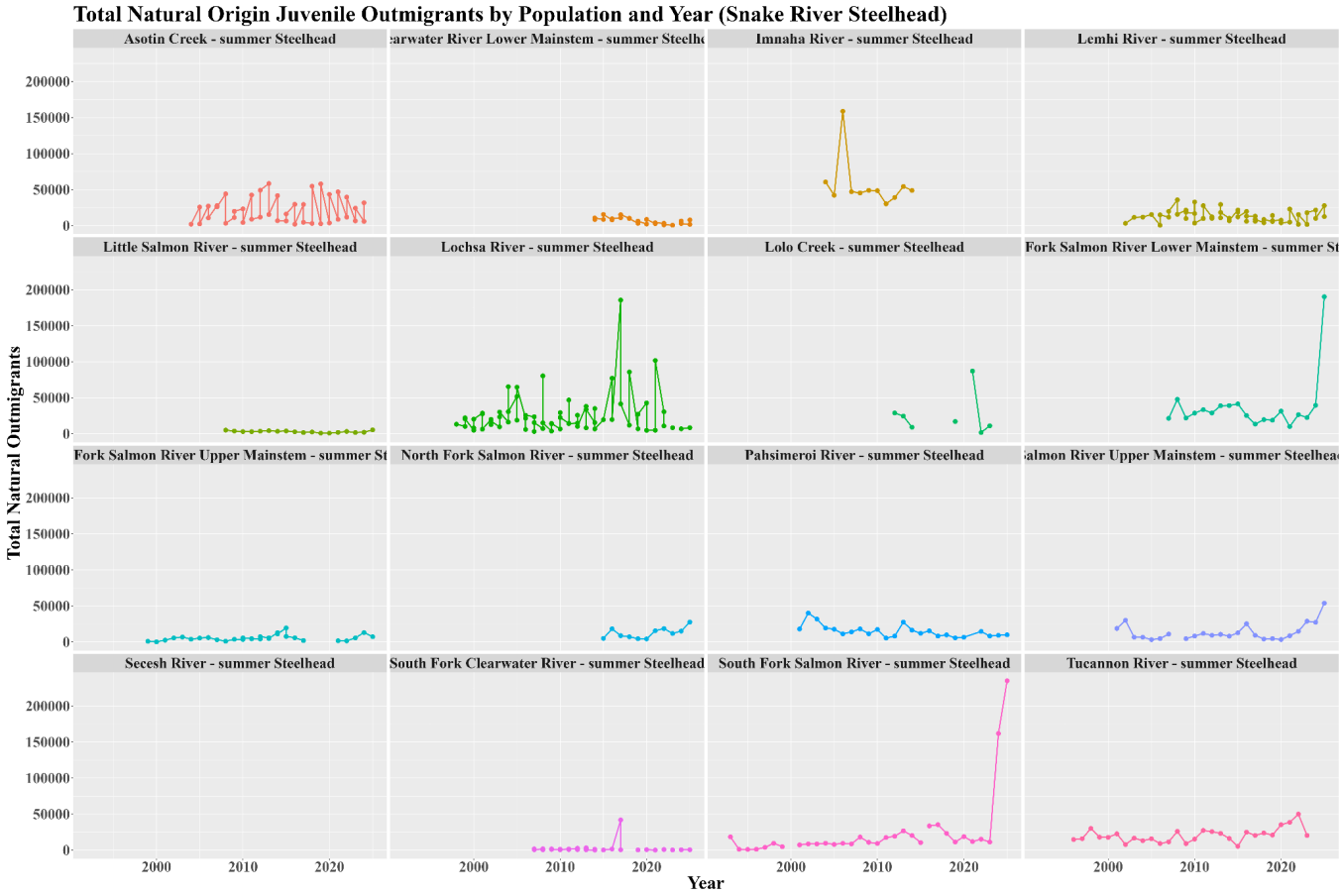


Figure 26: A line graph showing total natural origin juvenile outmigrants by year for each population in the Snake River Steelhead subset, faceted by population.

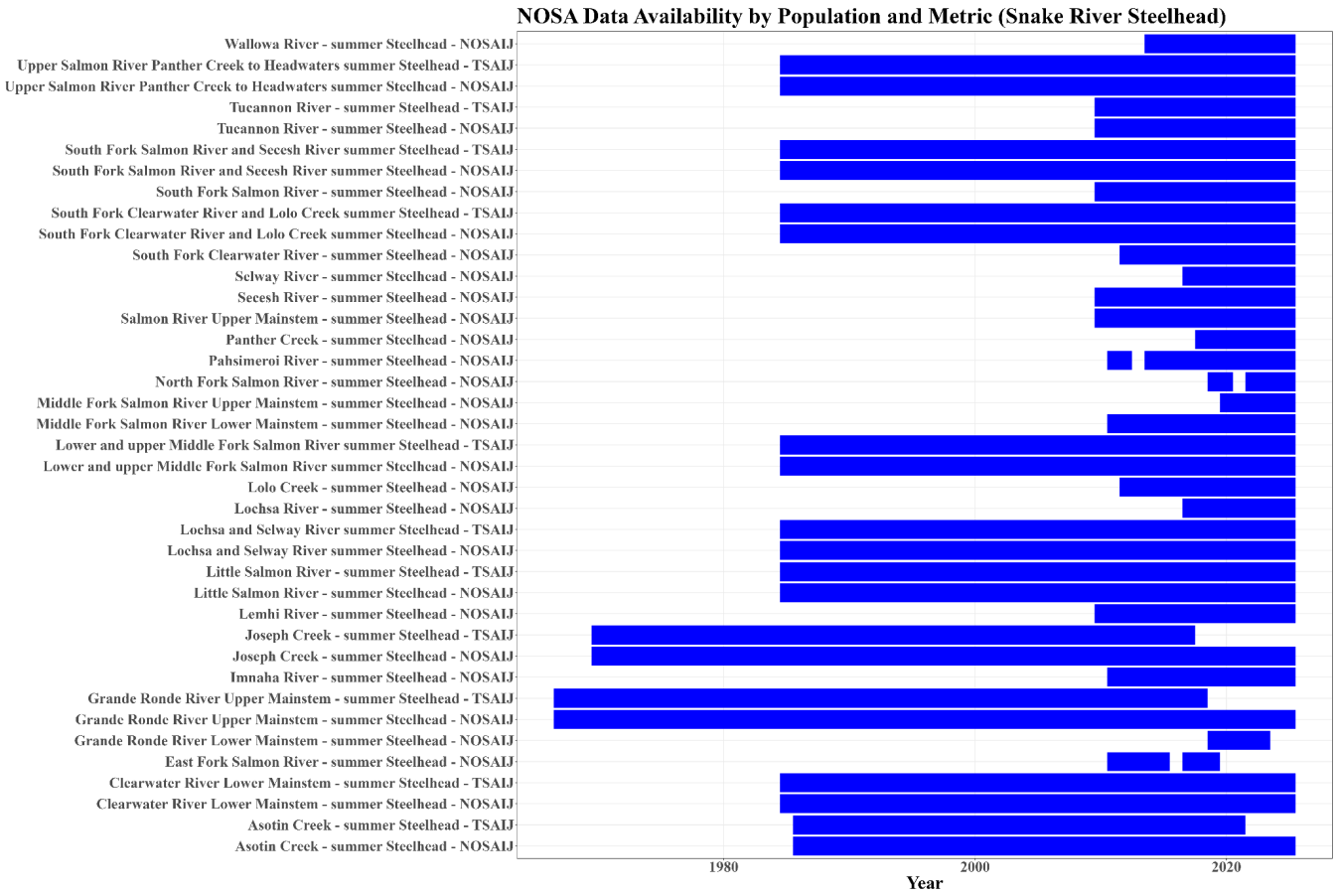


Figure 27: A bar graph showing data availability for StreamNet NOSA records by population and metric for Snake River Steelhead, including total spawner abundance including jacks (TSAIJ) and naturally occurring spawner abundance including jacks (NOSAIJ).

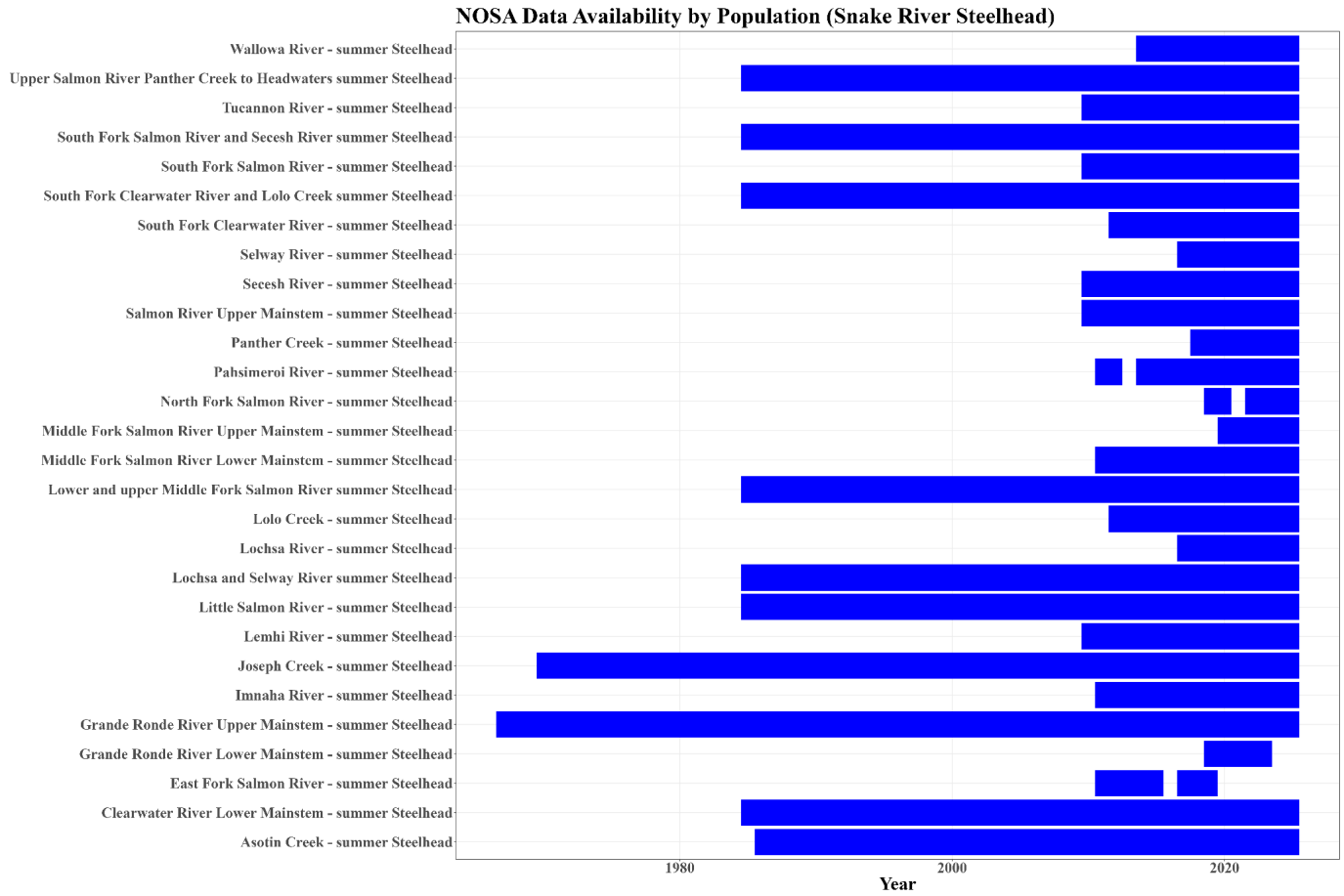


Figure 28: A bar graph showing data availability for StreamNet NOSA records by population for Snake River Steelhead.

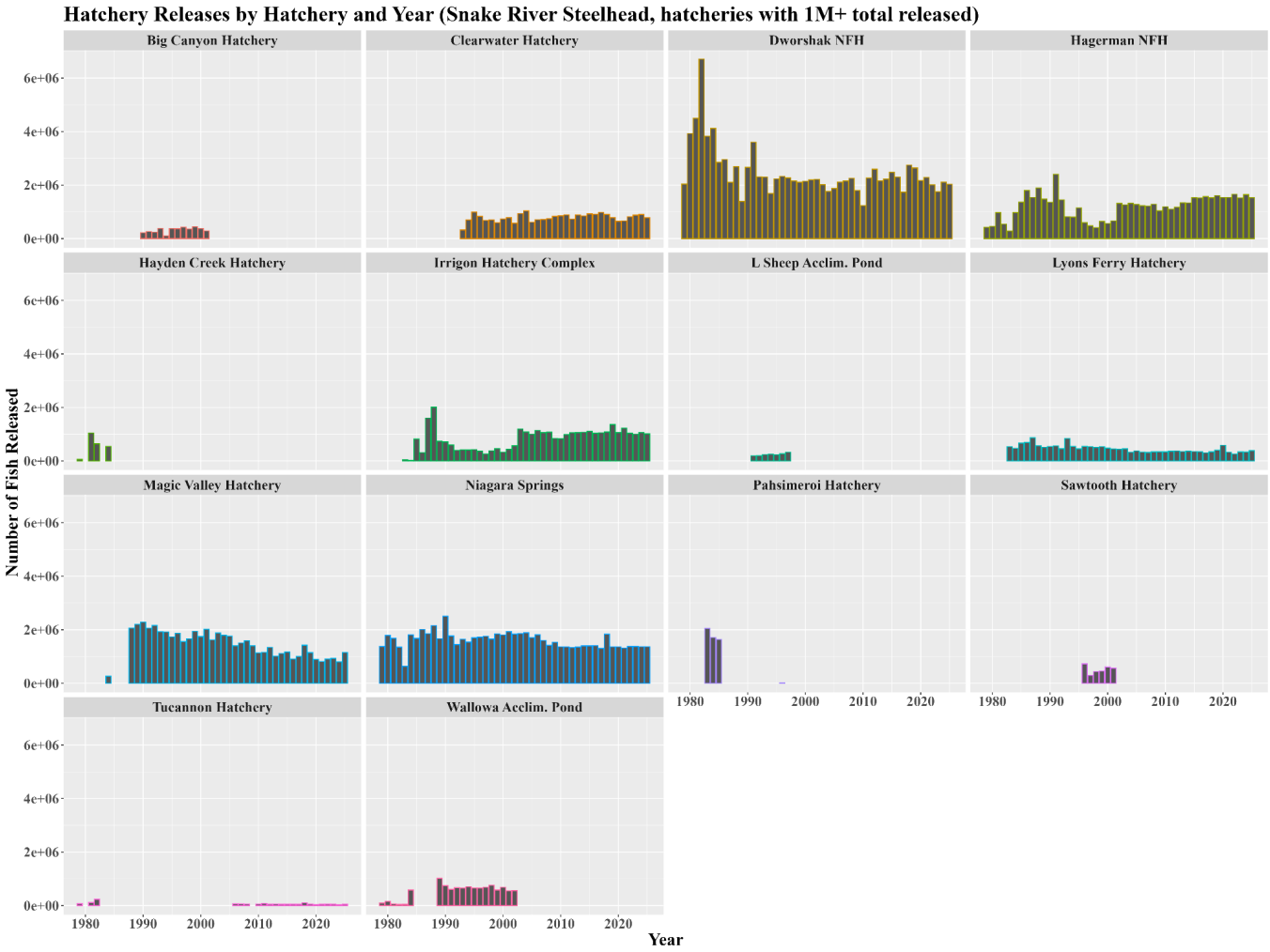


Figure 29: A bar graph showing the number of hatchery origin fish released by year for each hatchery in the Snake River Steelhead subset, faceted by hatchery. Only hatcheries with more than 1 million total fish released across all years are shown.

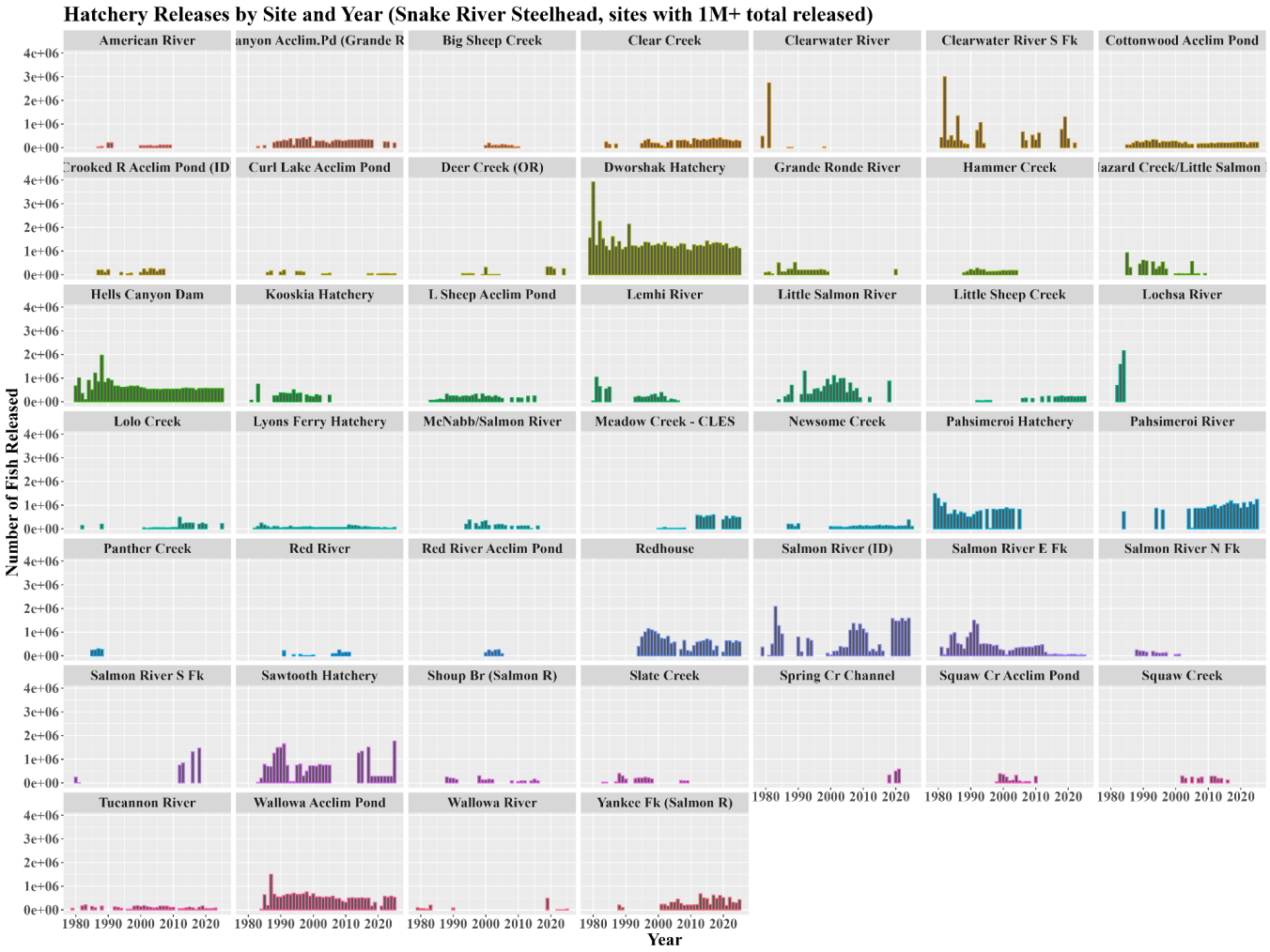


Figure 30: A bar graph showing the number of hatchery origin fish released by year for each release site in the Snake River Steelhead subset, faceted by site. Only sites with more than 1 million total fish released across all years are shown.

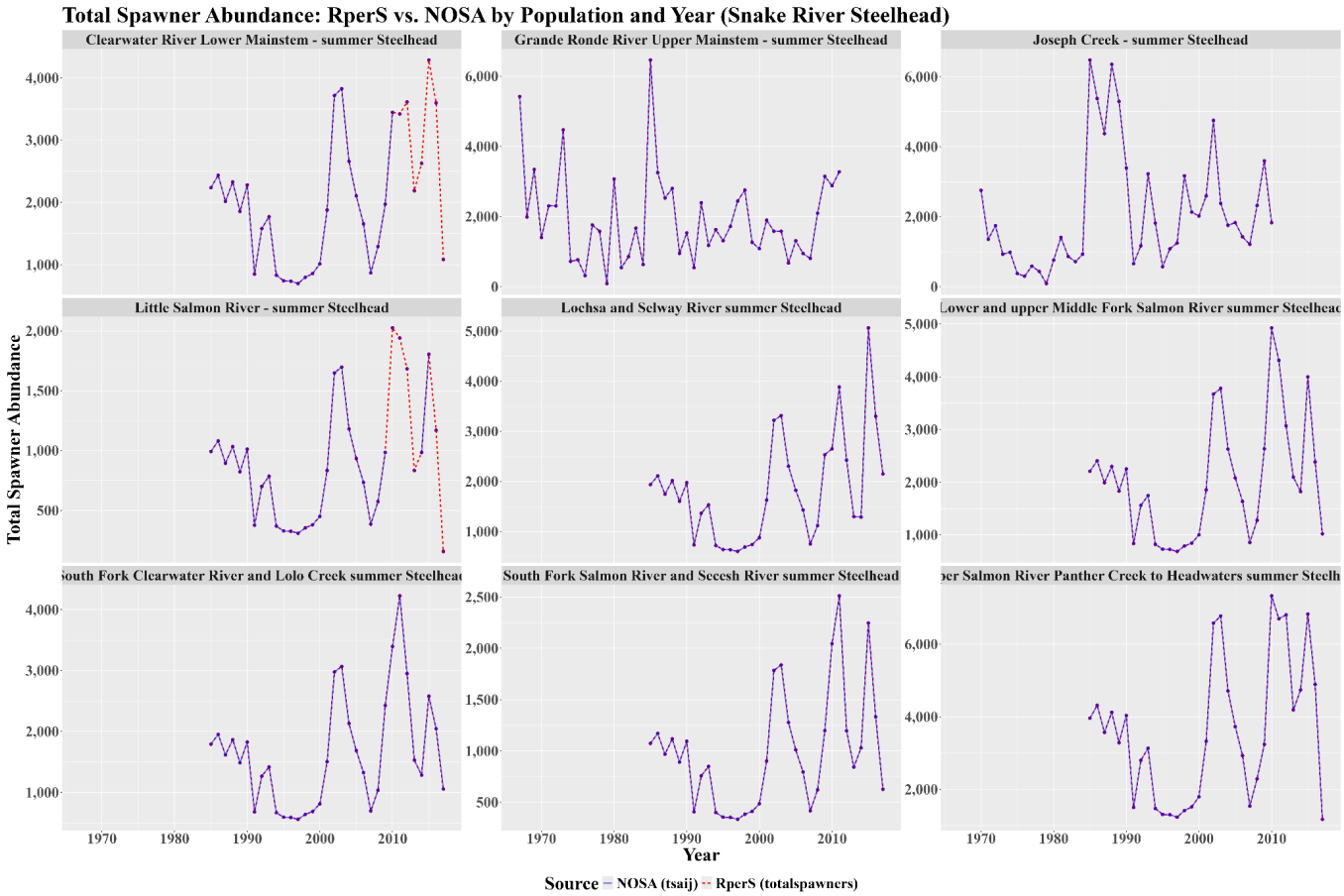


Figure 31: A line graph showing total spawner sundance by year for Snake River Steelhead populations, faceted by population, comparing estimates from NOSA (blue solid line) and RperS (red dashed line).